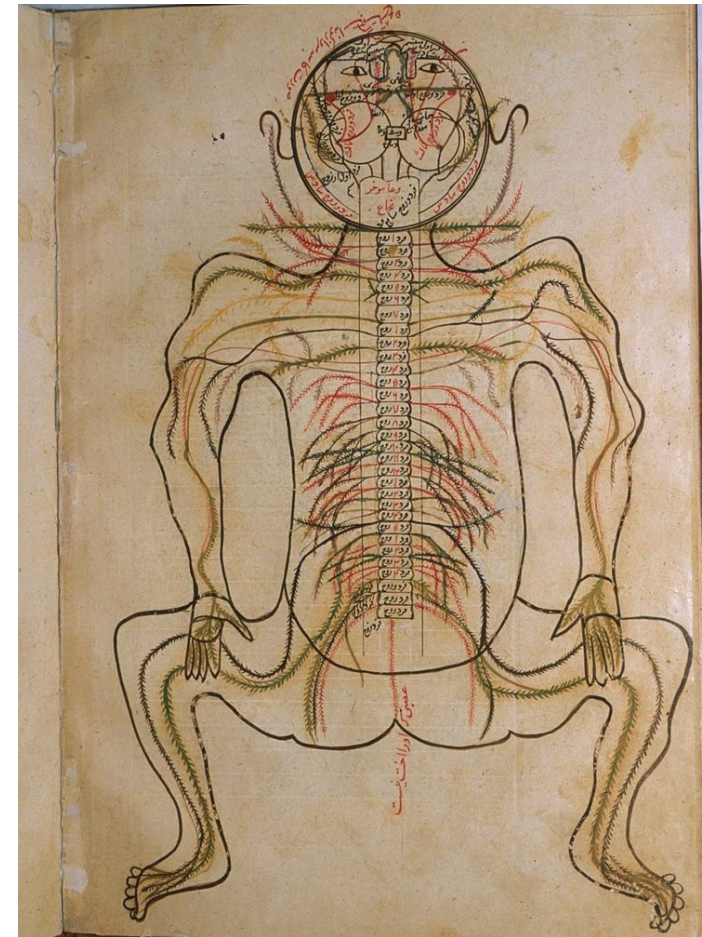


Proprioception Interoception and Exteroception



*Anatomical illustration of the nervous system
by Mansour Ibn-Ilyas, ca. 1400*



University of Trento

Agenda

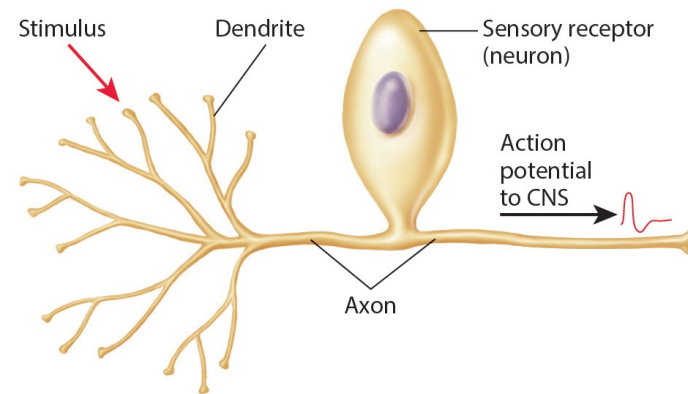
- Introduction to sensory receptors
- Proprioception:
 - Position sense → balance
 - Kinesthesia
 - Spindle fibers
 - Golgi organ
- Interoception:
 - Example of the carotid bodies and the control of respiration
- Exteroception:
 - Sensory receptors
 - Cutaneous sensory system
 - Gustation - olfaction
 - Hearing
 - Vision

Sensory receptors

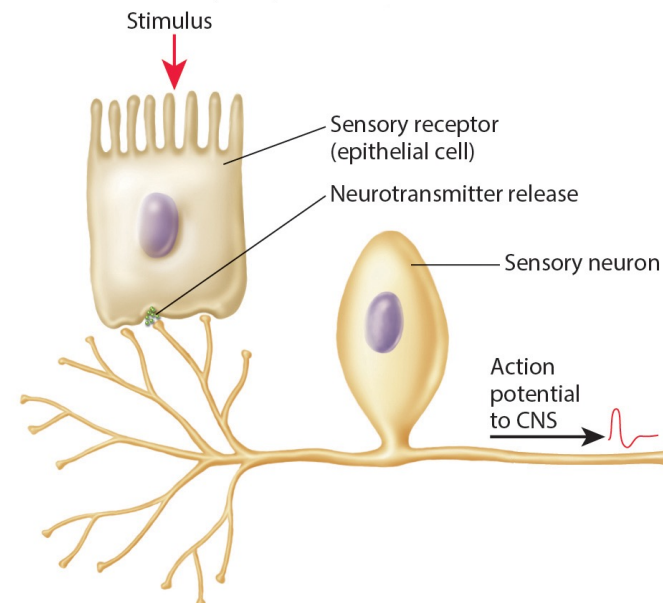
- Sensory systems **convert stimuli from body or external environment** to a signal that changes **membrane potential** of sensory neurons

- **Sensory receptors are:**

- **Neurons or specialized epithelial cells**
- **Recognize stimulus** and initiate **graded potentials** in itself or adjacent cell
- When the **response is strong enough**, an **action potential** is sent to the **CNS**



(a) A neuron as a sensory receptor

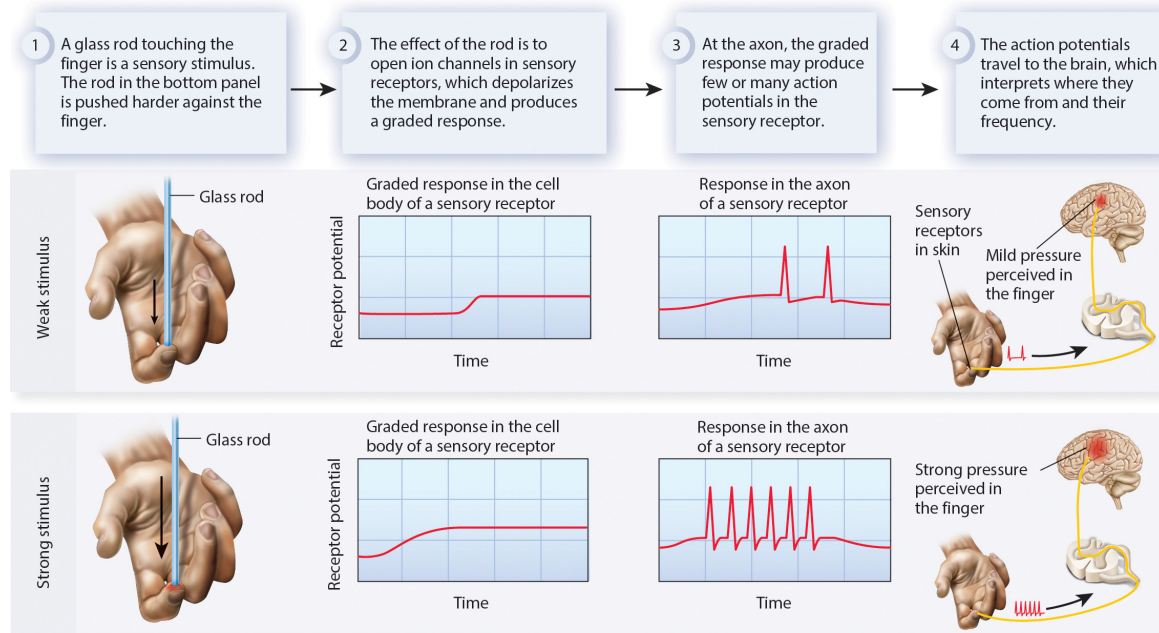


(b) A specialized epithelial cell as a sensory receptor

Strength and type of stimulus

- Strength of the stimulus is indicated by the **frequency of action potentials** generated:

—Strong stimulus generates many action potentials in a short period of time

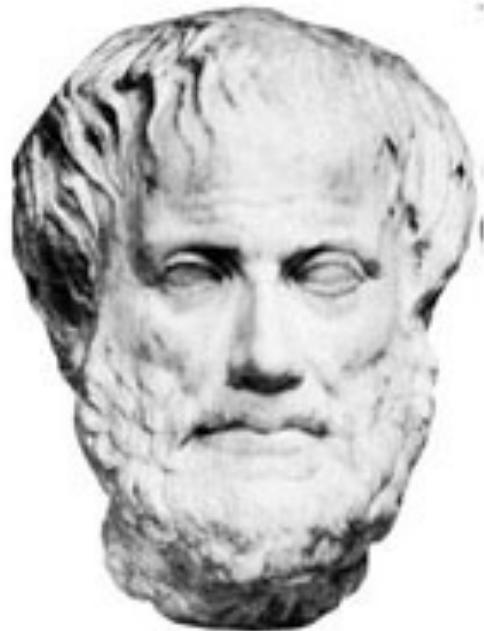


- Brain interprets higher frequency of action potentials as a **more intense stimulus**
- Different stimuli produce different sensations because they **activate specific neural pathways** dedicated to processing only that type of stimulus

Categories of sensation

Aristotle's Five Senses

De Anima (4th c. BCE)



Vision
Audition
Olfaction
Gustation
Touch

Categories of sensation



The Sensory Modalities

Sherrington (1906)

Exteroception

- Distance Senses
 - Vision
 - Audition
- Chemical Senses
 - Gustation
 - Olfaction
- Skin (Cutaneous) Senses
 - Touch (Tactile)
 - Temperature (Thermal)
 - Pain (Nociception)

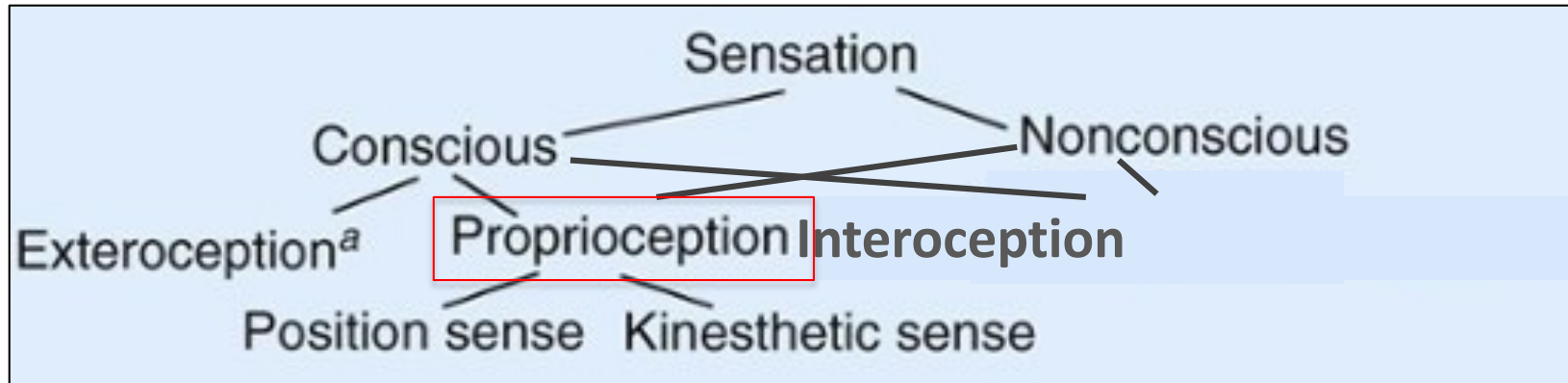
Proprioception

- Kinesthesia
- Equilibrium (Vestibular)

Interoception

- The sense of the physiological condition of the body

Proprioception and consciousness



PROPRIOCEPTION: from *proprioceptor*, from Latin *proprius* "own" + reception. Coined in 1906, by English neurophysiologist C.S. Sherrington (1857-1952).

POSITION SENSE: equilibrium (vestibular)

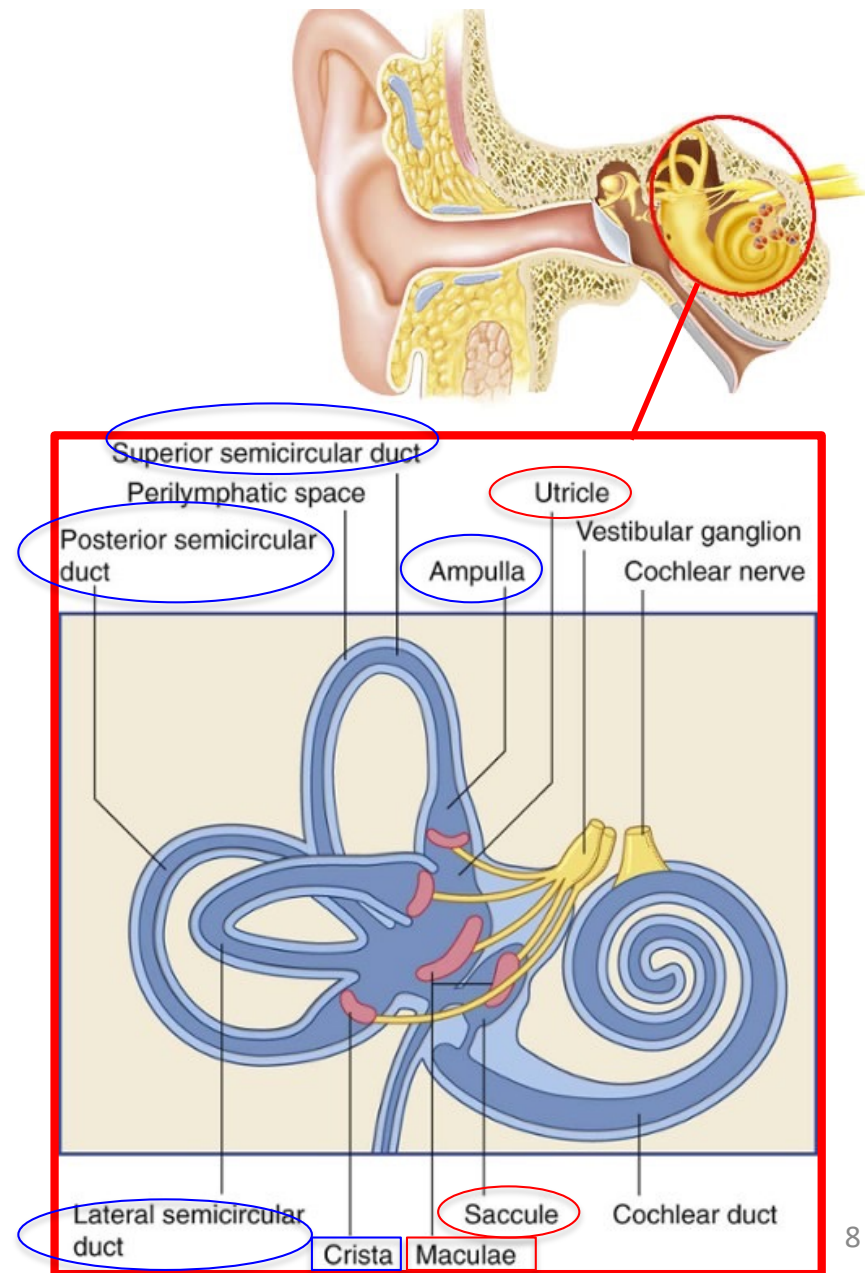
KINESTHESIS: the ability to sense the orientation and movement of the body and its parts

Proprioception: Balance

Vestibulo-chochlear nerve innervates the **vestibular system** in the **inner ear**:

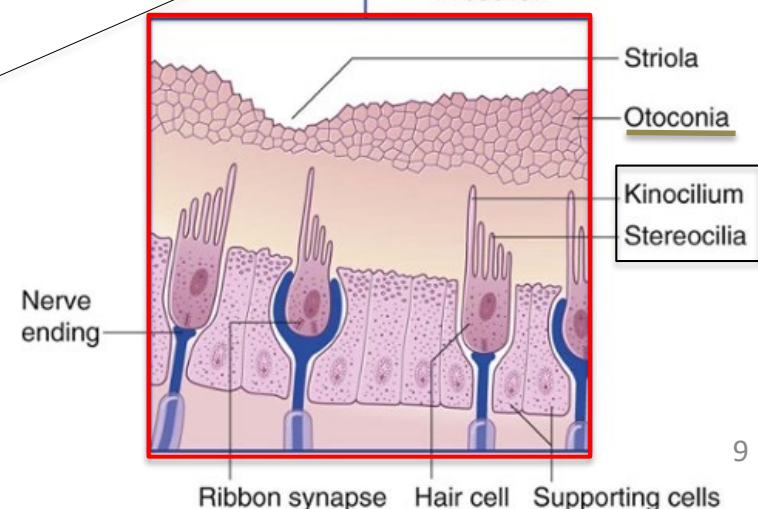
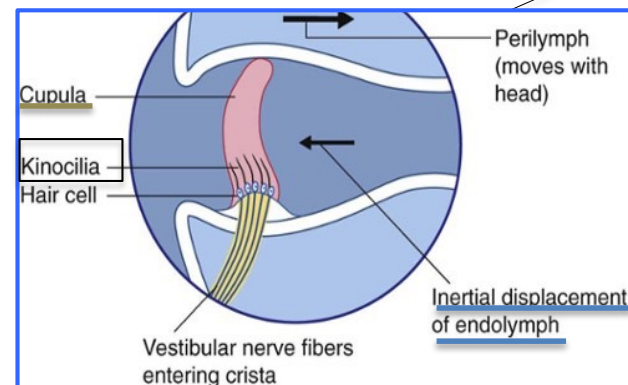
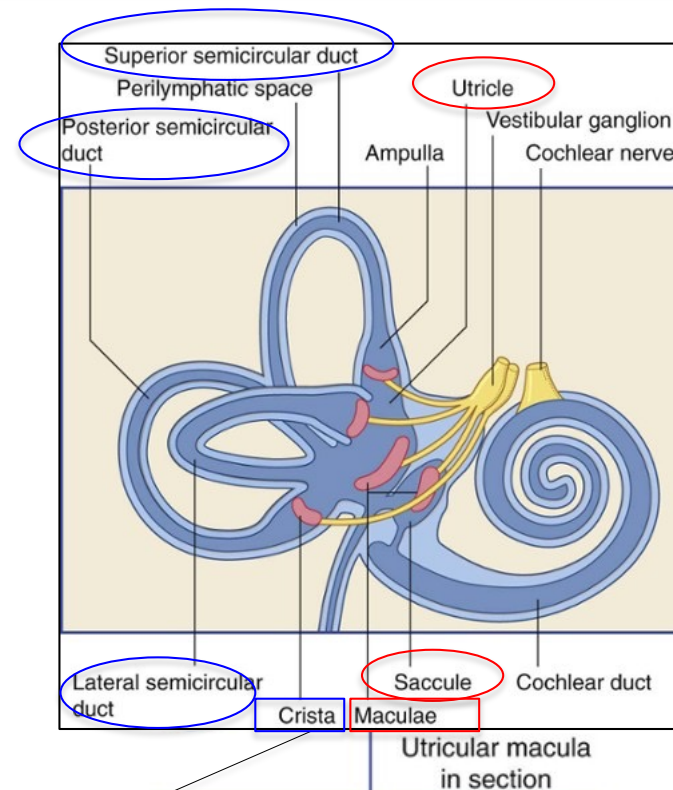
- **Dynamic labirinth** (signals head movement): **3 semicircular canals ending with an ampulla** housing a **crista** → response to **angular acceleration**
- **Static labirinth** (signals head position): **Utricle (~horizontal) and saccule (~vertical) each containing a macula**: response to **linear acceleration** (i.e. **gravity, vertical plane or walking, horizontal plane**)

→ control of balance, coordination and movement of head and neck.



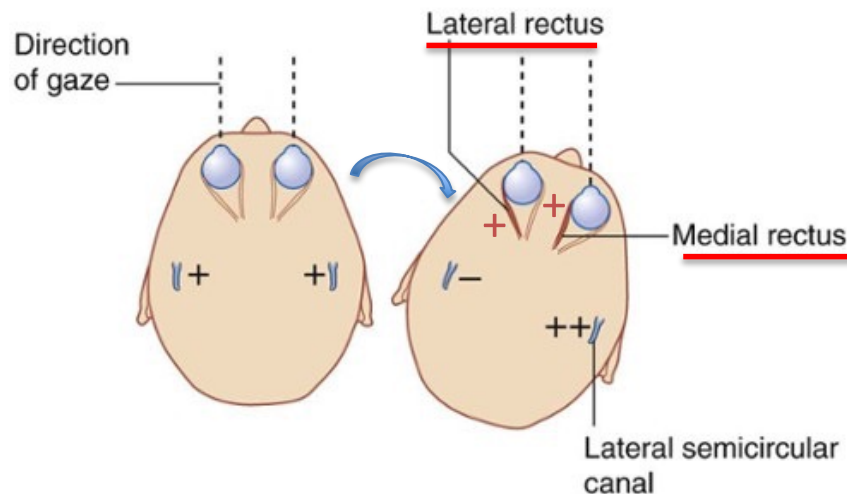
Cristae and maculae

- **Sense organs are bathed in endolymph**
- **Ampullar cristae** contain **hair cells** that sense when **long uniformly distributed kinocilia** penetrating in a gelatinous structure called cupula are **inertially displaced** by ampullar **endolymph**
- **Utricular and saccular maculae** contain **hair cells** that sense through **stereocilia (~100/cell)** and **kinocilium (longer 1/cell)** a differential displacement exerted by otoconia (a gelatinous matrix in macula)

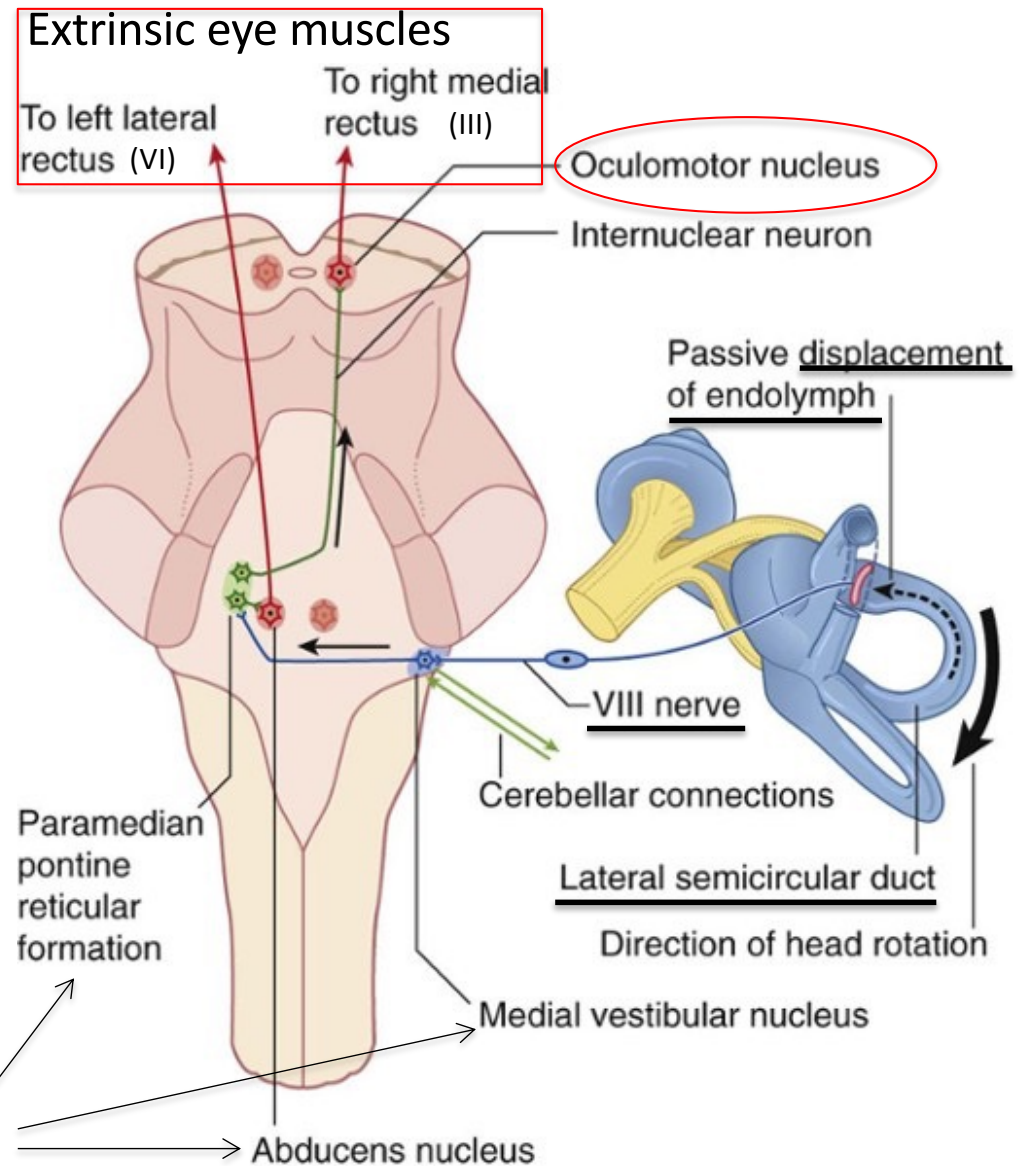


Vestibulo-cochlear (VIII) nerve projects to the brain stem

VESTIBULO-OCULAR REFLEX

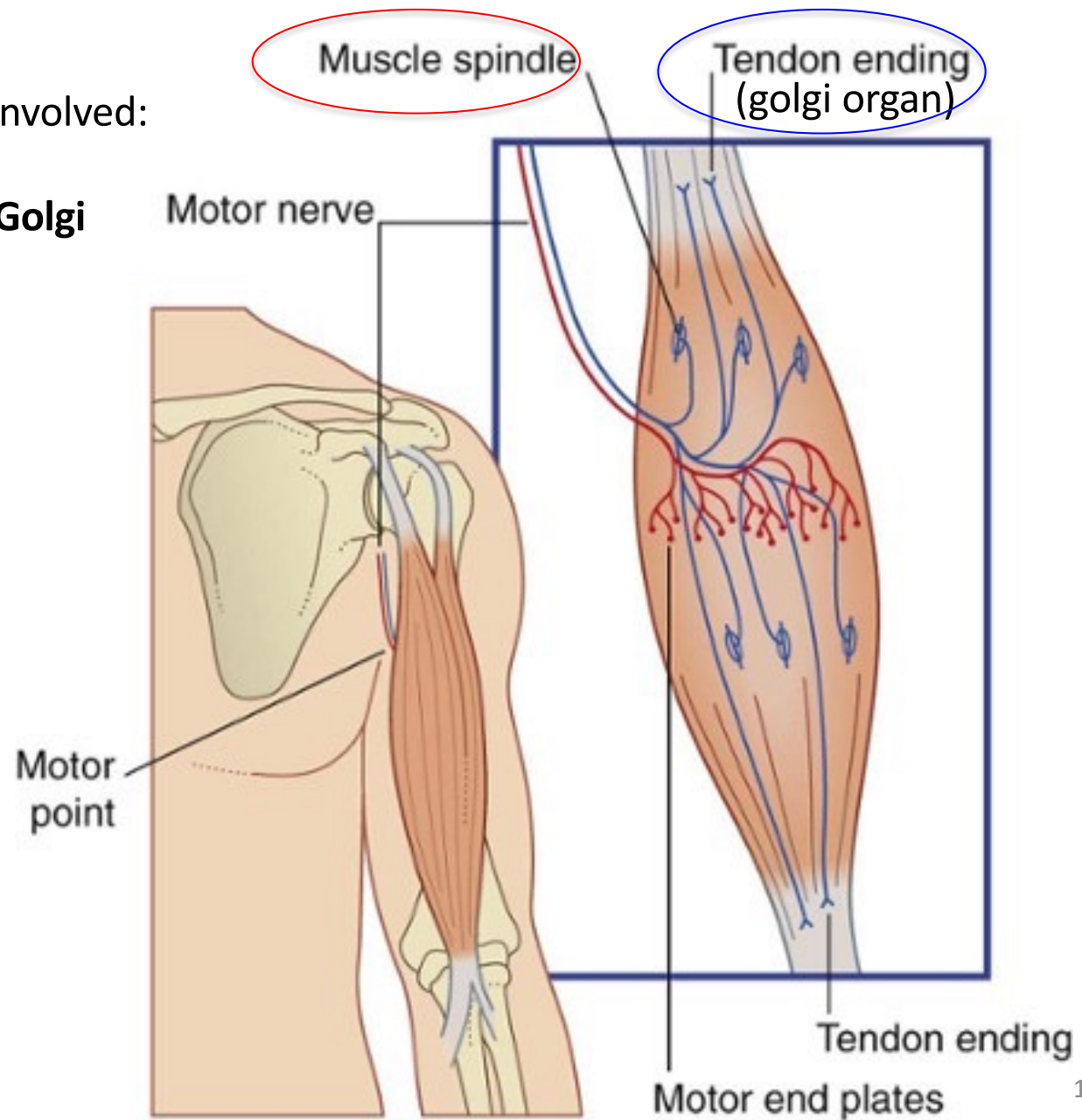


→ coordination between movements of head and neck and the control of eyes muscles



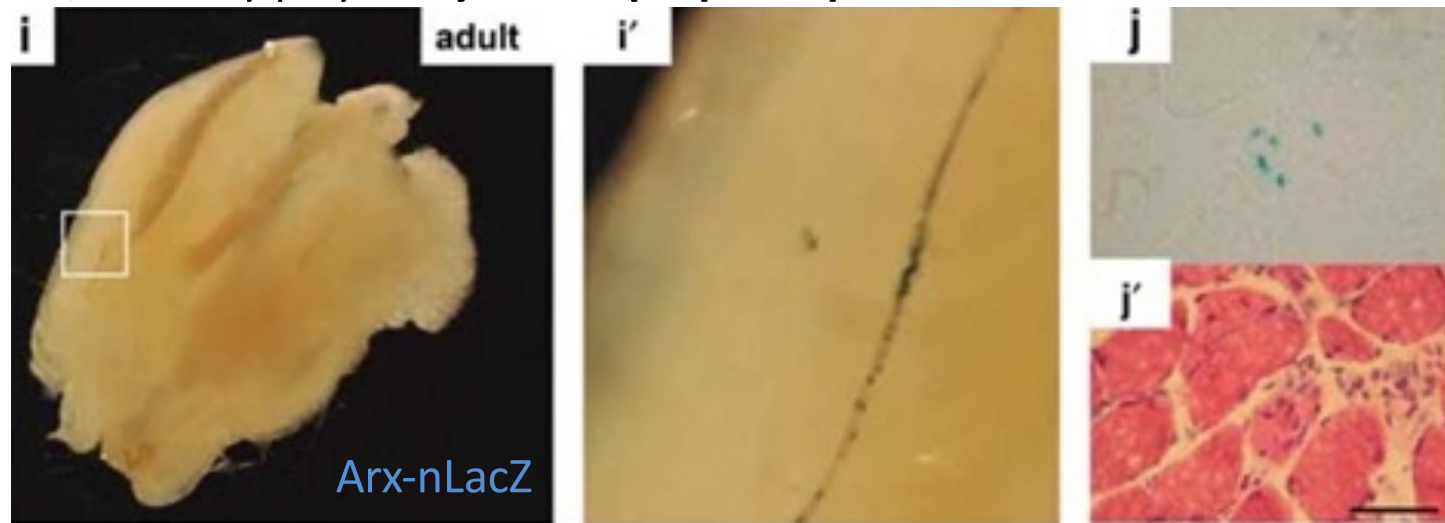
Proprioception: **Kinesthesia**

- Two major structures involved:
 - Muscle spindles**
 - Tendon organ of Golgi**

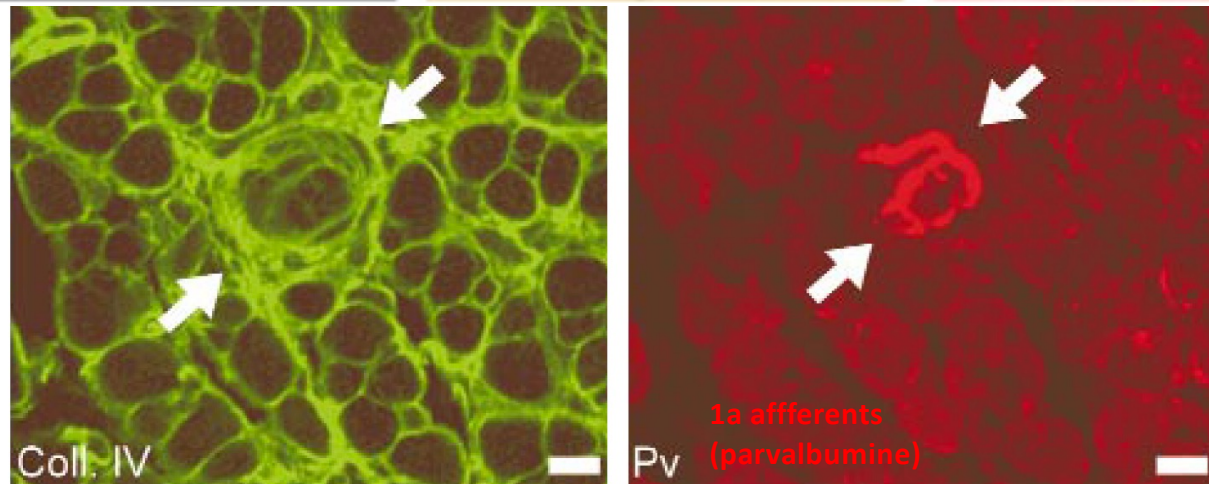


Spindle/intrafusal fibers

- Skeletal muscle consist of **two functionally very distinct** classes of **fibers**:
 - **Extrafusal**
 - **Intrafusal (spindle)**
- Contribution of **spindle fibers** (small and rare) to **overall muscle tension is negligible**, but they play a **key role in proprioception**



Biressi 2008



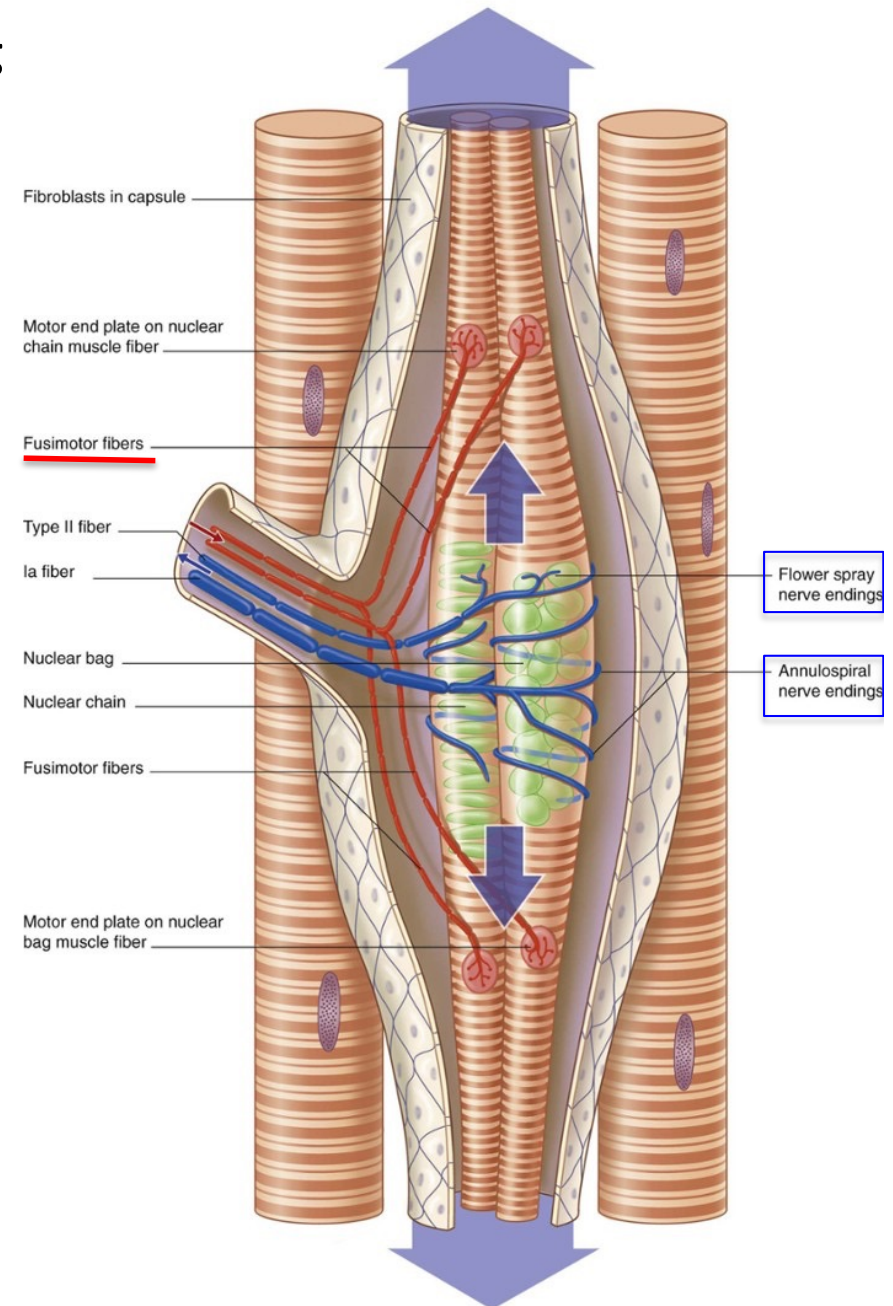
Tourtelotte 1998

Classification of mammalian axons

Diameter/Conduction Velocity Properties			Functional
Fiber Types	Fiber Diameter (μm)	Conduction Velocity (m/sec)	
Aα	12-22	70-100	Erlanger & Gasser Numerical A α — Somatic Motor — Ia — Proprioception — Ib — Muscle Spindle — Ia — Golgi Tendon Organ — Ib β — Touch, Pressure, Proprioception — II γ — Motor to Muscle Spindles δ — Pain (fast), Cold, Touch — III B — Preganglionic Sympathetic C — Dorsal root — Pain ^{slow}, Hot, other receptors (mechano), reflex — IV — Sympathetic — Postganglionic Most Numerous Diameter and Velocity DECREASE MN Simplified by DrAyushGoel @ www.MedicoNotebook.com
Aβ	5-13	30-70	
Aγ	3-8	15-40	
Aδ	1-5	12-30	
B	1-3	3-15	
C (unmyelinated)	0.2-1.2	0.2-2.0	

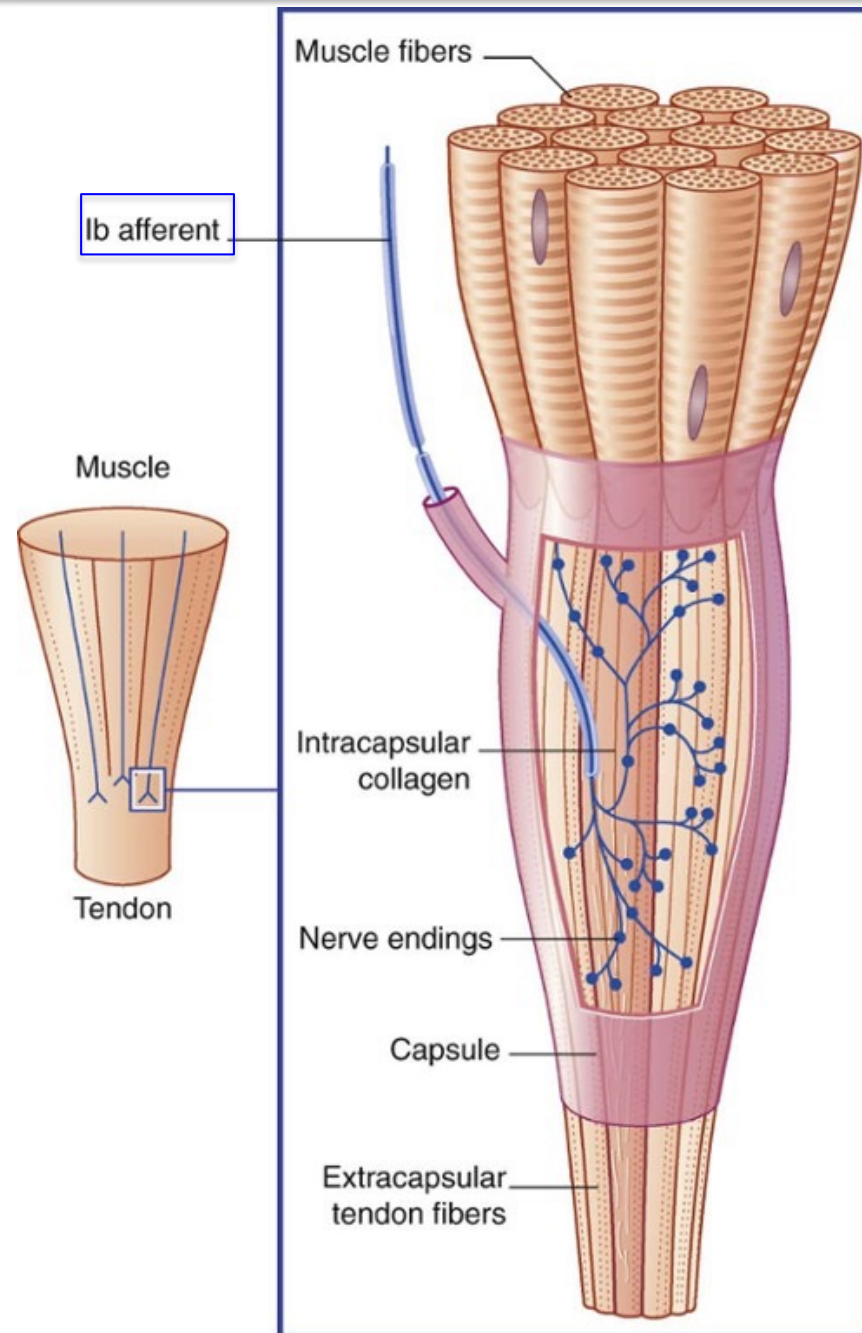
Spindle/intrafusal fibers: **stretch detectors**

- Spindle fibers are **specialized structures ~1cm long**
- Their abundance vary **from a few dozens to hundreds in different muscles**
- Particularly **abundant in slow muscles** such as:
 - **Back muscles**
 - **Muscles of the neck**
 - **Intrinsic muscles of the hand**
- Both **motor** and **sensory** innervation
 - **Fusimotor**: myelinated **A γ** fibers
 - **Sensory**:
 - **type Ia sensory neurons** form **annulospiral wrappings around the equator** of the spindles and sense the **rate of muscle stretch**.
 - **type II neurons** form **flower spray ends** and sense the **degree of muscle stretch that is maintained**.



Golgi organs: **force detectors**

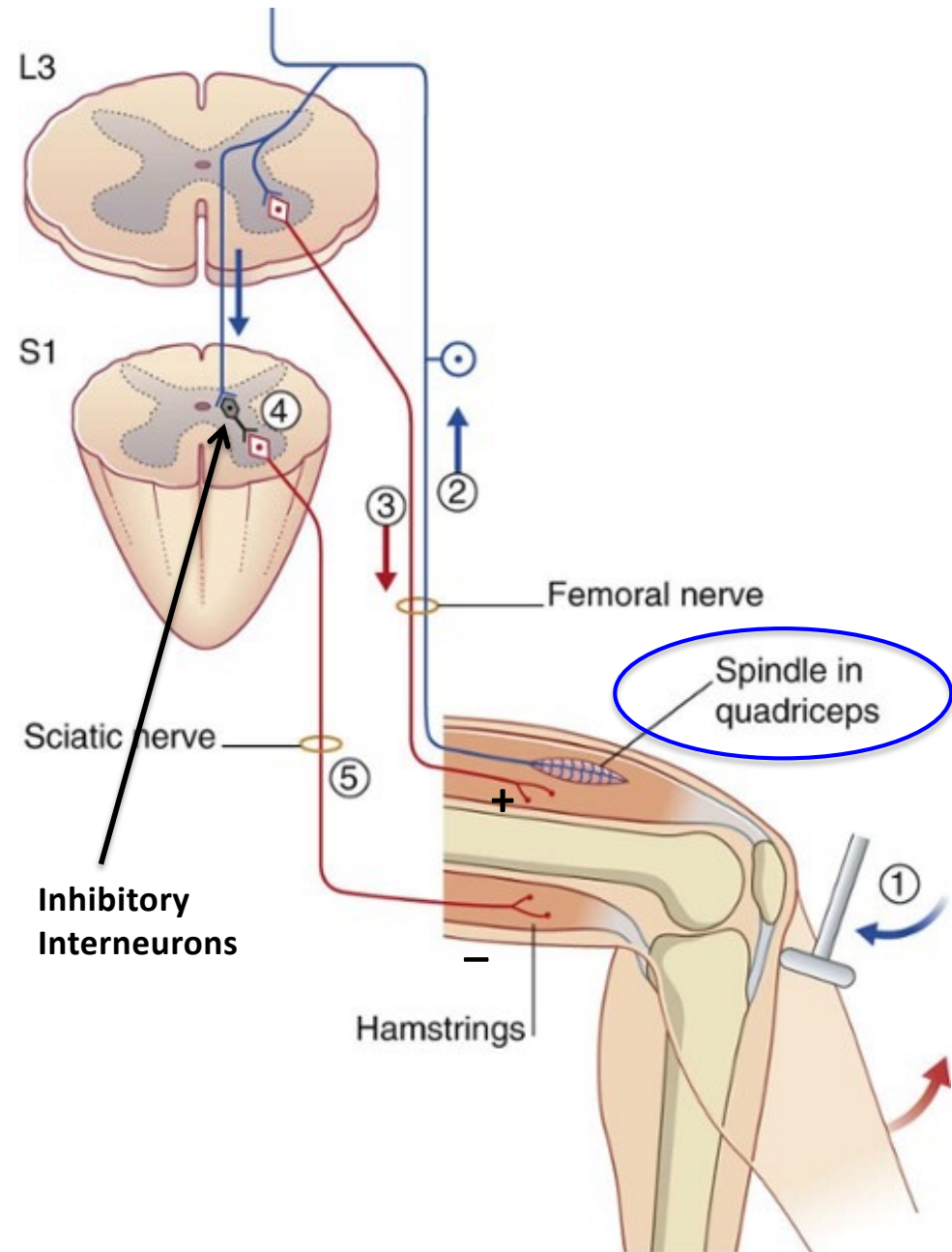
- Golgi organs are at the **muscle-tendon junction**
- **Particularly sensitive to forces** developed by **contracting muscle fibers** → important in the response to **active movements**



Passive stretch and spindle fibers

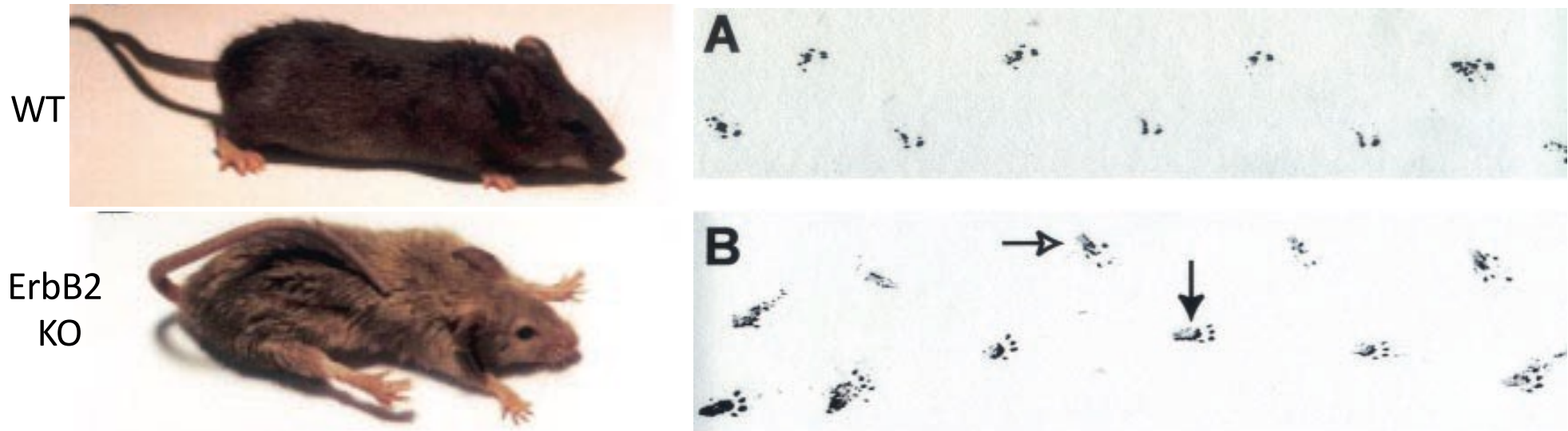
- **Passive stretch** of muscle spindles occurs when the **entire muscle is passively lengthened**
- **Passive stretch** may activate stretch reflexes **like the patellar reflex:**
 1. A tap of the **patellar ligament** stretch the spindles of the quadriceps
 2. The **spindle discharged excitatory impulses** to the spinal cord (**afferents**)
 3. A **monosynaptic** relay activates **motorneurons** innervating the **quadriceps** (homonymous) and inducing **extrafusal fiber contraction**
 4. Through **inhibitory spinal interneurons** the activity of extensor muscle (quadriceps) is coordinated with that of **flexor muscles** (→ **hamstring muscle is relaxed**)

This control is called **RECIPROCAL INHIBITION**



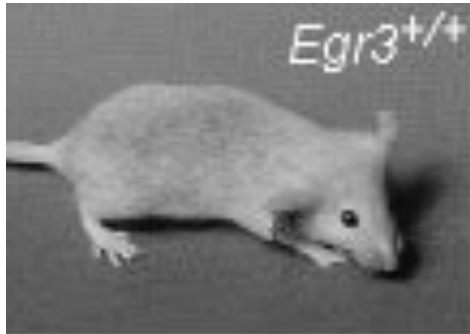
Muscle-specific ErbB2 KO mice have impaired proprioception and gait

- **Proprioception is impaired** in muscle-specific ErbB2 KO mice
- This is resulting in severe **gait alterations**



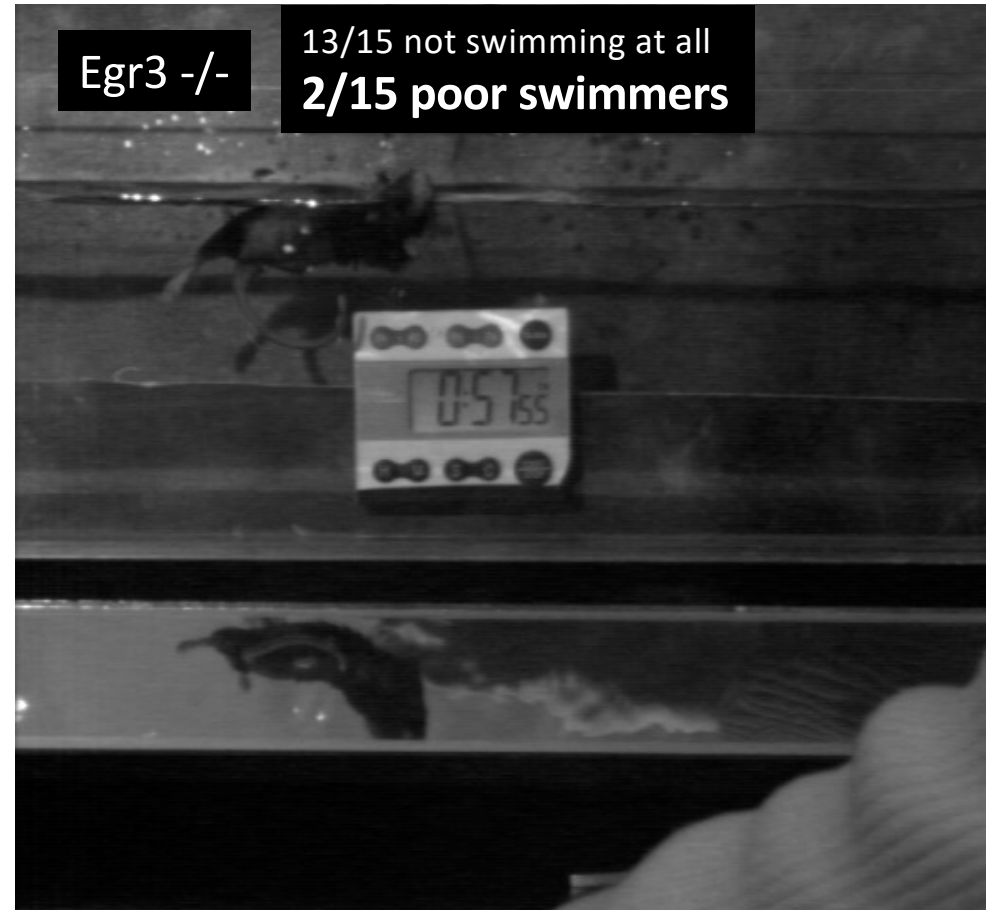
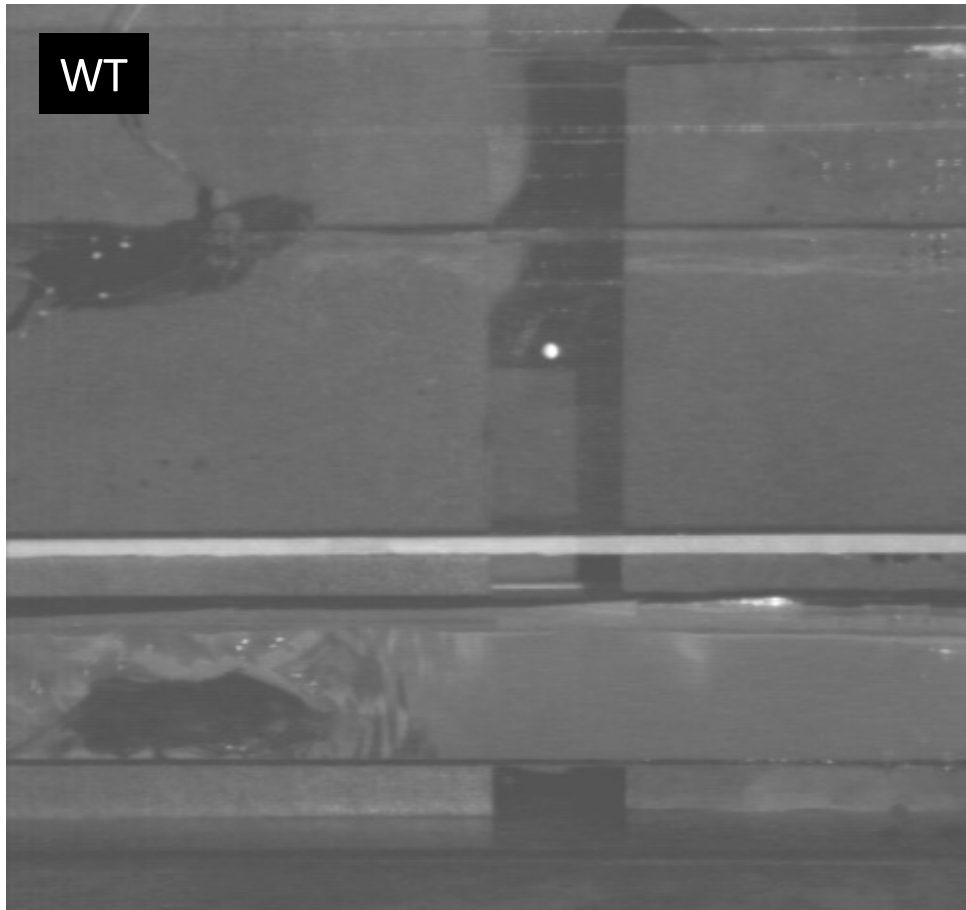
Andrechek et al. 2002

Defective locomotor pattern in absence of proprioception in Egr3 KO



Tourtelotte 1998

Akay 2014



Defective mouse locomotor pattern in absence of proprioception

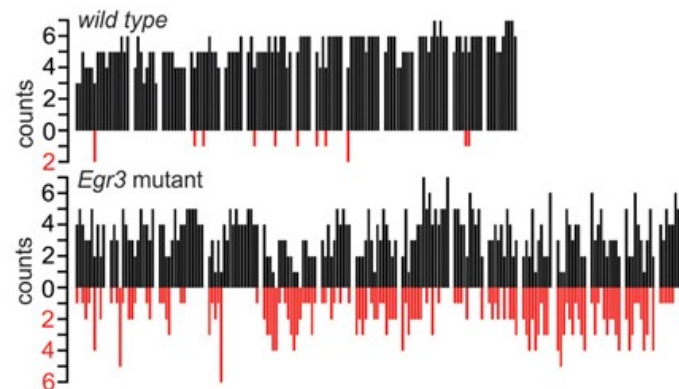
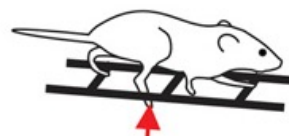
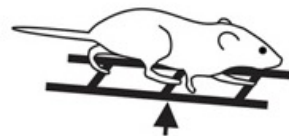
WT



Egr3 -/-



Walking on an horizontal ladder

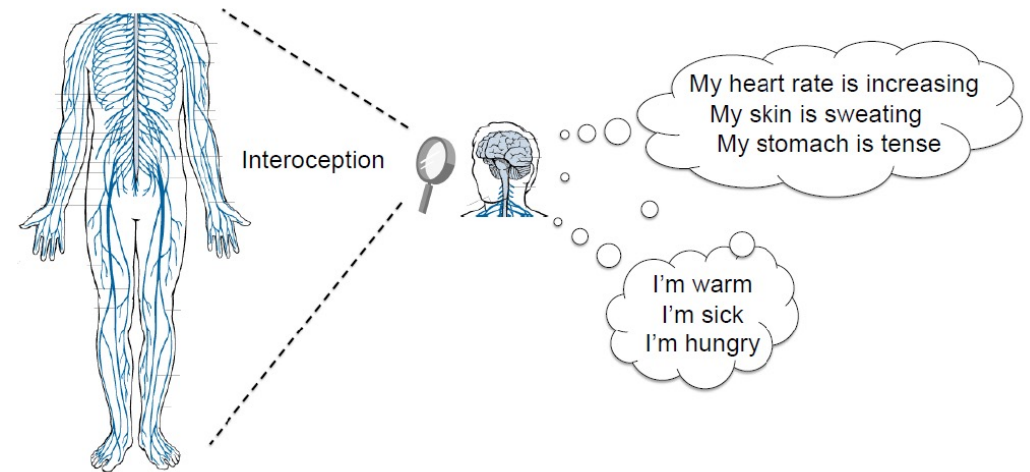


Interoception

Interoception: the sense of the physiological condition of the body

Interoception refer to the ability of human body to perceive feelings such:

- **Visceral sensation**
- **Vasomotor activity**
- **Muscular sensation (metabolic activity)**
- **Pain**
- **Temperature**
- **Hunger**
- **Thirst**
- **“Air hunger” (hypoxia, CO2 levels)**



- Interoception depends on afferent neurons of different types conducting **mechanical, thermal, chemical, hormonal and metabolic information**
→ play a **key role in controlling homeostasis** (i.e. kidney function, heart function, gastrointestinal function, metabolism, inflammation, neuroendocrine function etc)

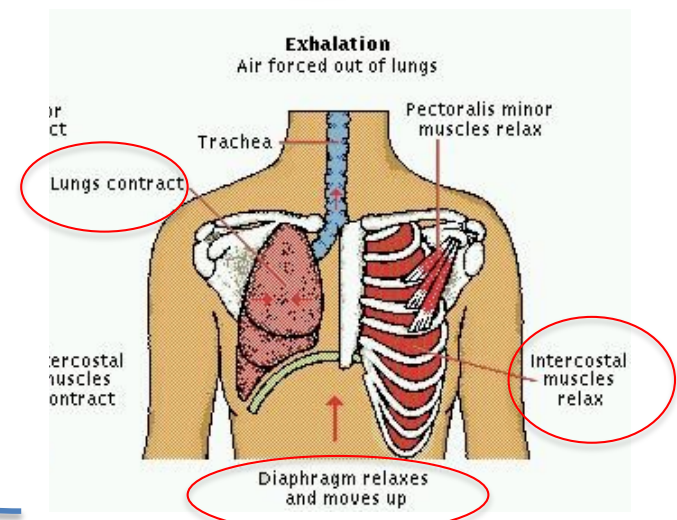
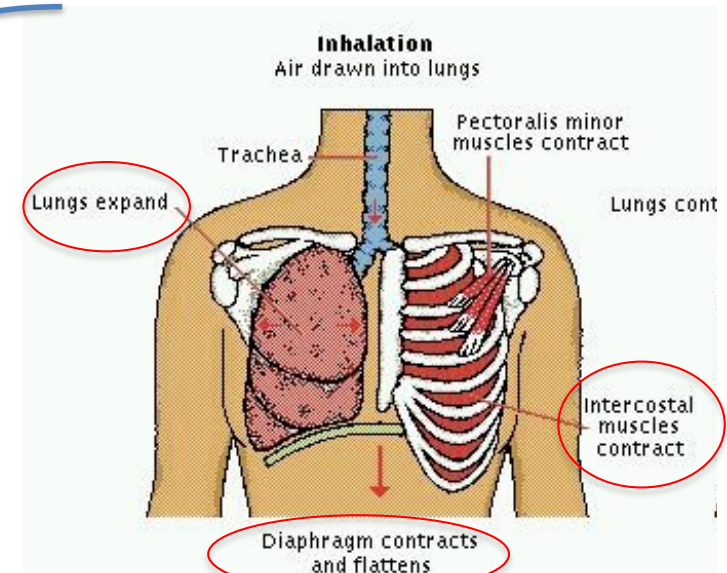
Interoception and the control of respiration

Efferent pathway:

- Nerve fiber from respiratory center leaves the brain and descend in anterior part of lateral column of spinal cord.
- These nerve fibers terminate in the motor neurons in the anterior horn cells of the cervical and thoracic segments of spinal cord.
- From motor neurons two sets of nerve fiber arise which supplies particular muscle:
 1. Phrenic nerve fibers: supplies diaphragm
 2. The intercostal nerve fibers: supplies intercostal muscles.

Afferent pathway:

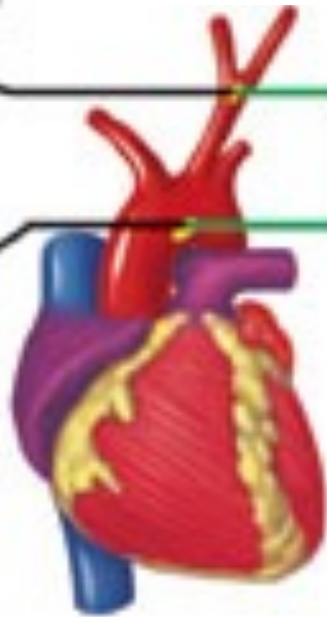
- From peripheral chemoreceptor and baroreceptor impulses are carried by glossopharyngeal and vagus nerves to respiratory center.



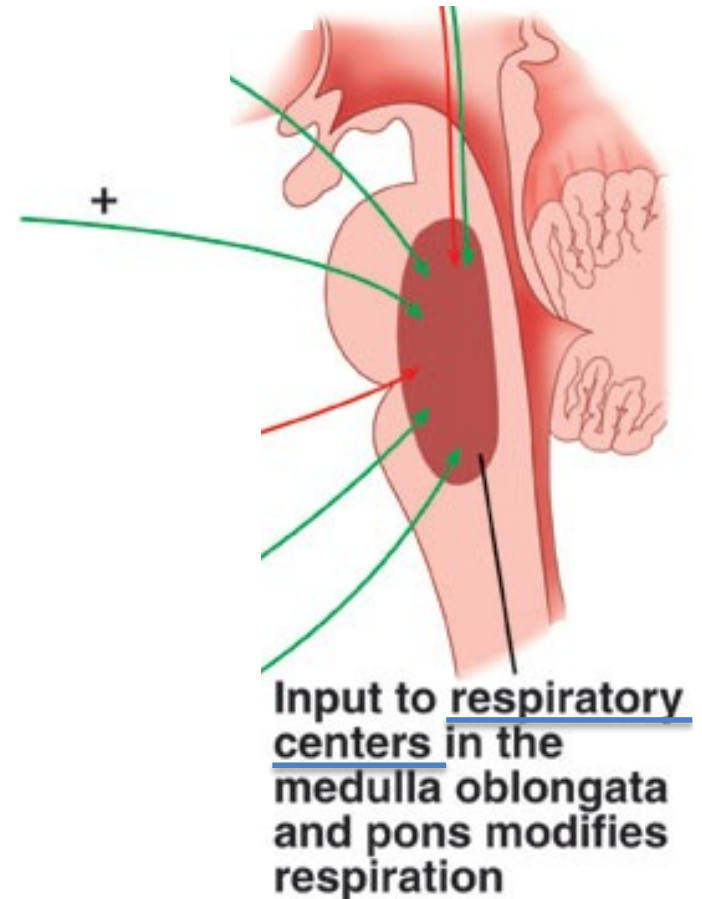
Afferent inputs from the peripheral chemoreceptors are activating brain stem respiratory centres

**Carotid
body**

**Aortic
body**



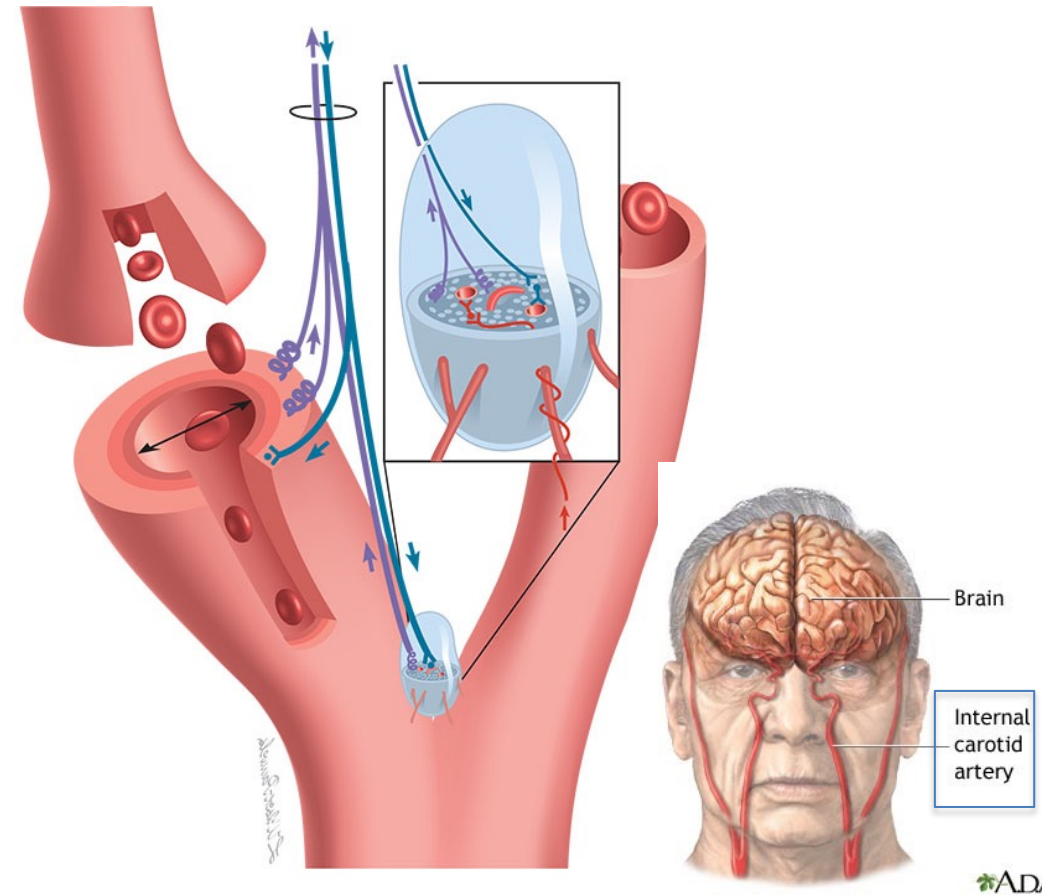
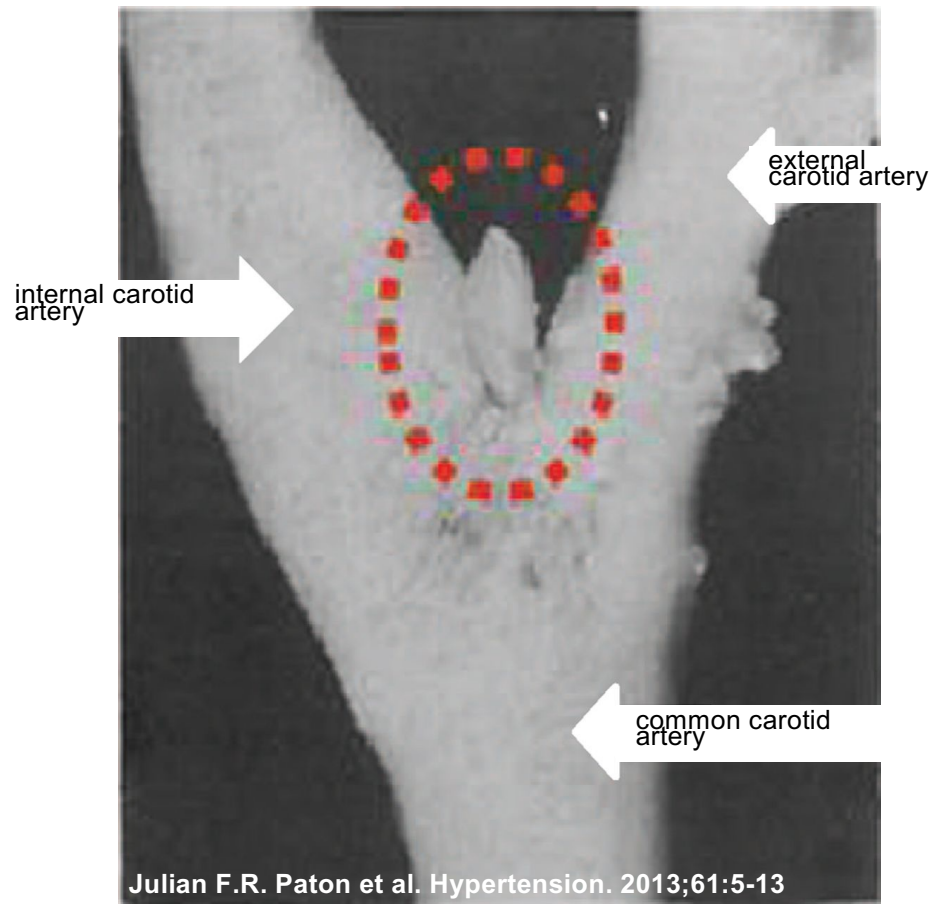
Carotid and
aortic body
chemoreceptors
 $\downarrow O_2$



Interoception: carotid body

Chemoreceptors important for the control of respiration are located in

- Internal Carotids (carotid body)
- Aortic Arches
- CNS



- The Carotid body is $\approx$ equivalent to the size of a **grain of rice**

Interoception: carotid body

- **Carotid bodies (CB)** respond to:
 - **Blood CO₂ levels/ protons (H⁺)**
 - **Blood levels of O₂**
 - **Hypoglycemia**
- **CB** are the **most perfused organ/** gram weight in the body (2000ml/min per 100mg tissue)

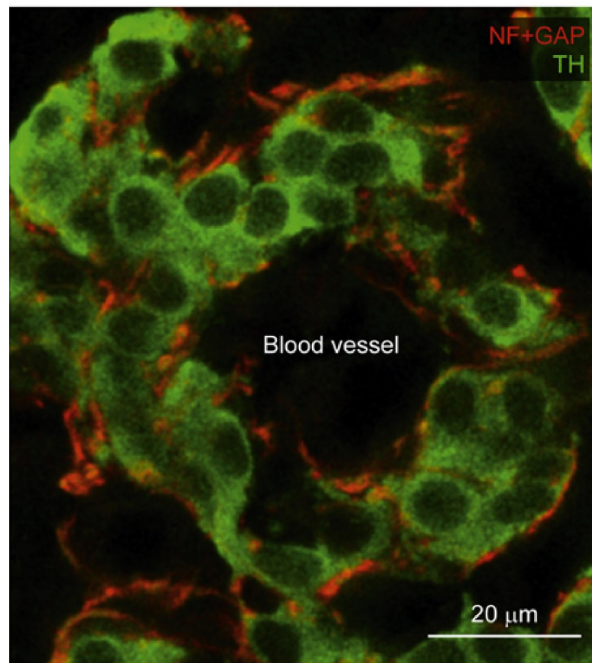


Drawing by De Castro

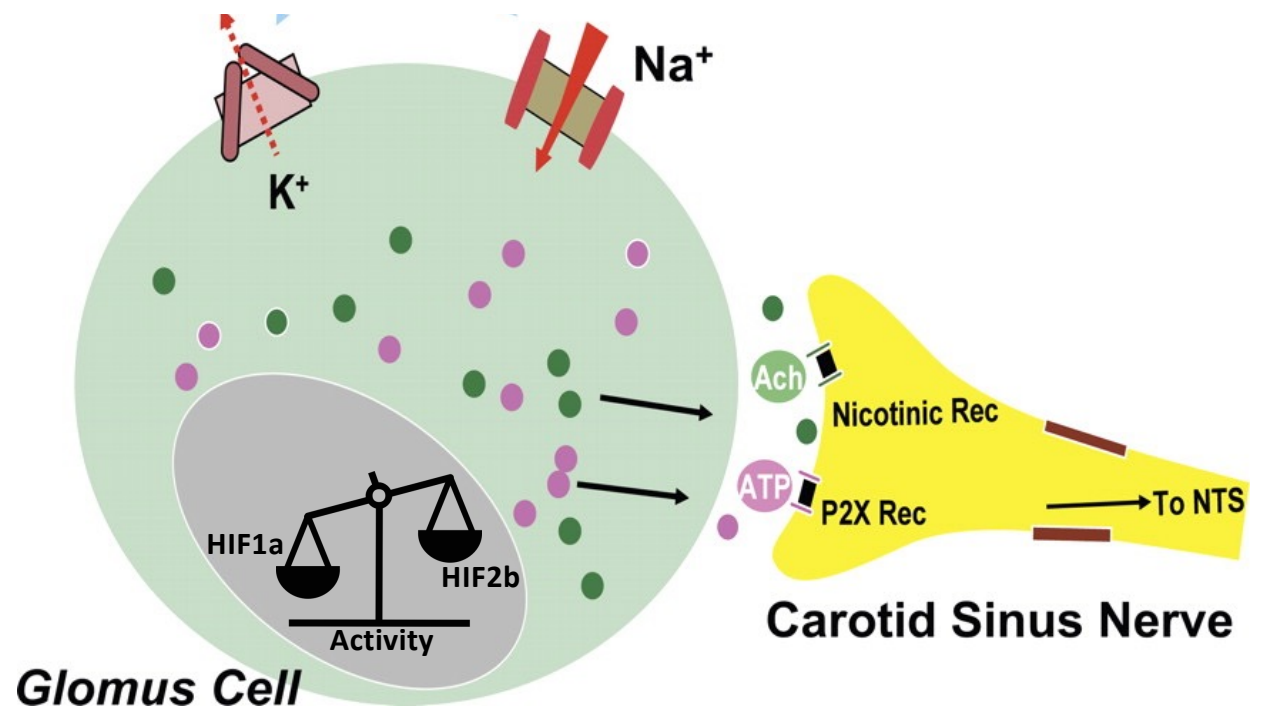
Glomus cells transduce the chemical stimuli to sensory terminal

Glomus cells are

- Sensitive to local changes in PO_2 and PCO_2 /pH

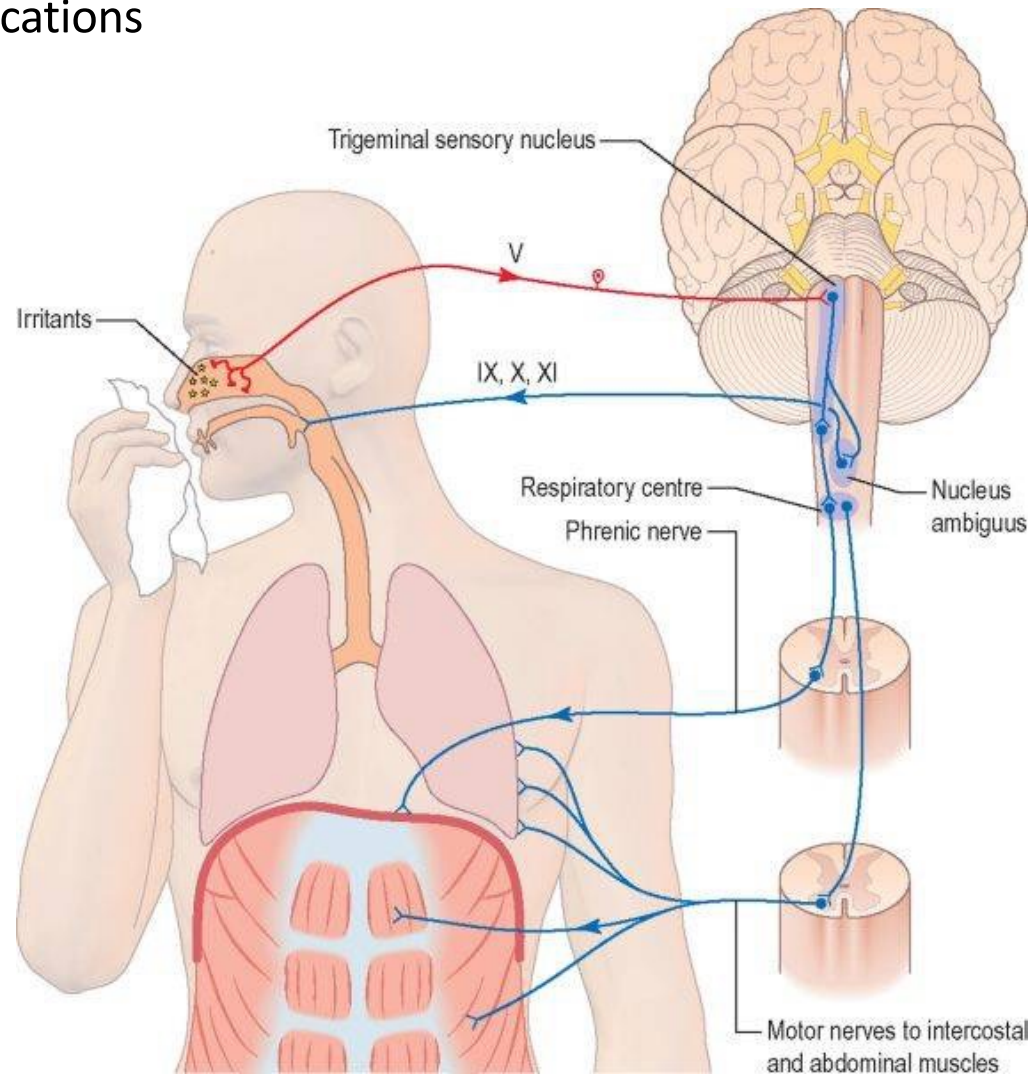


Neurofilament/GAP-43: sensory fibers
Tyrosine-hydroxylase (TH): Glomus cells



Coughing and Sneezing reflexes are influencing breathing

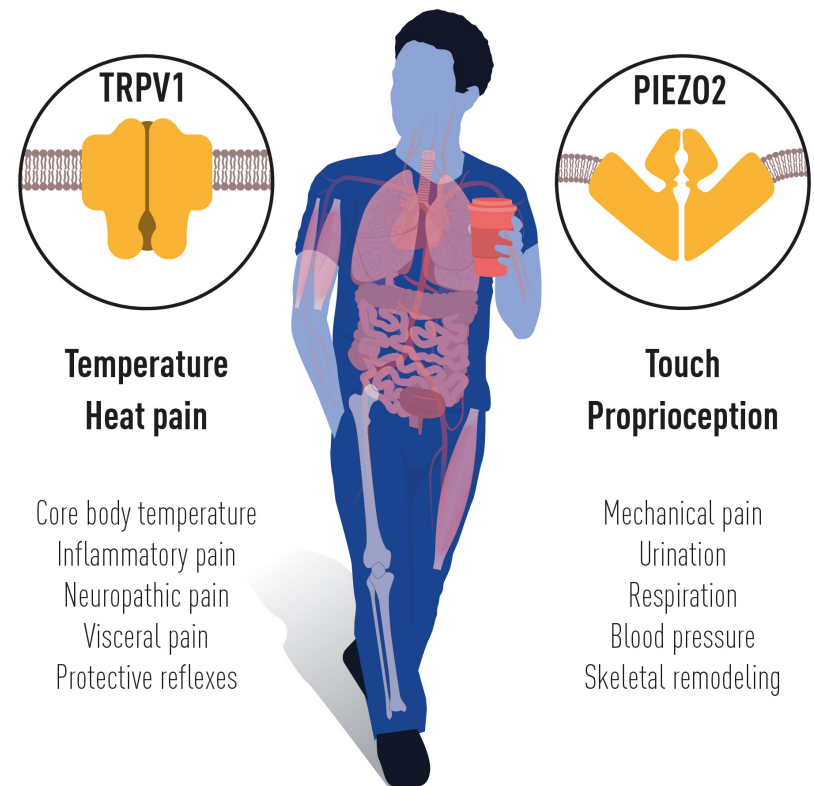
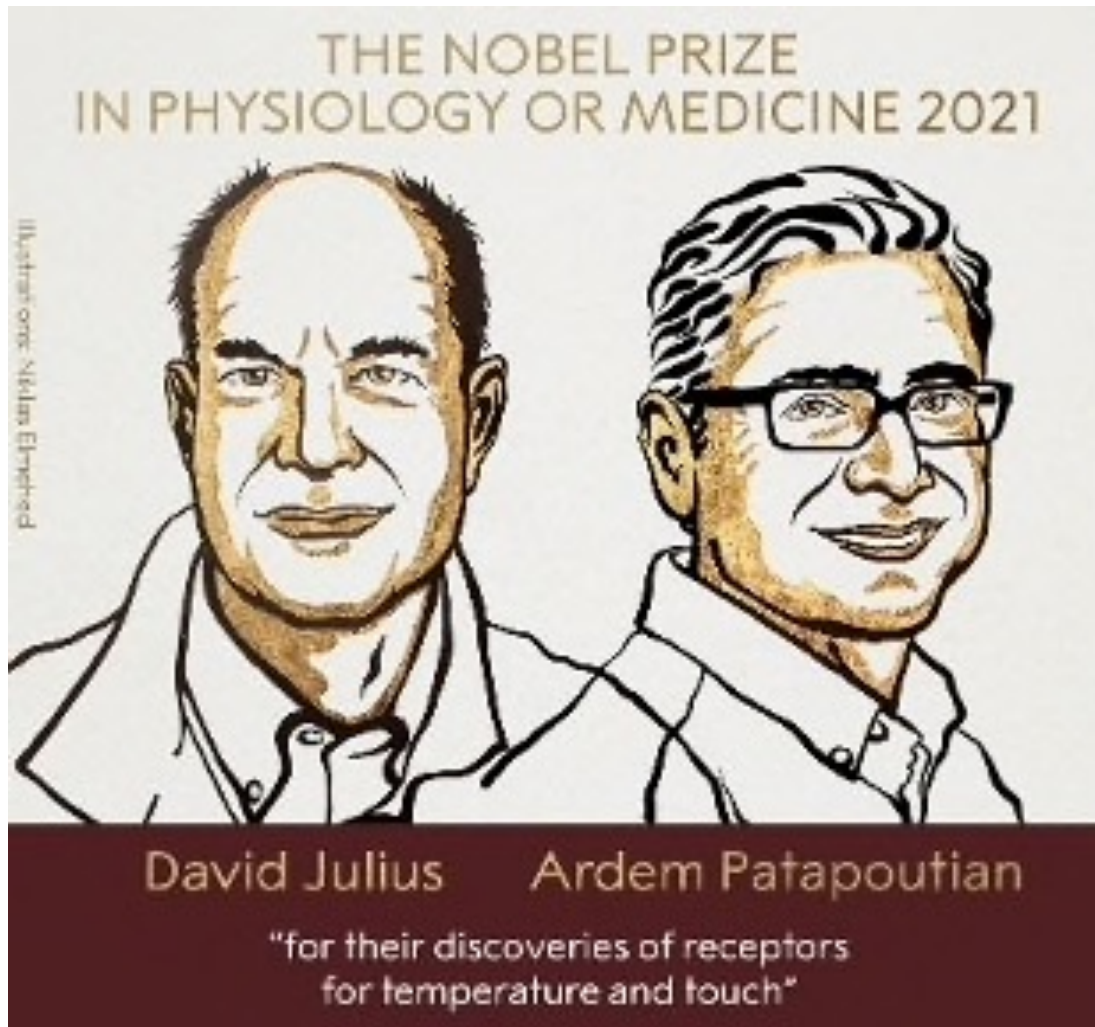
- Profound **changes in breathing** pattern is **associated with coughing and sneezing**
- **Cough and Sneezing** are **initiated by a wide variety of chemical and mechanical stimuli** at different locations



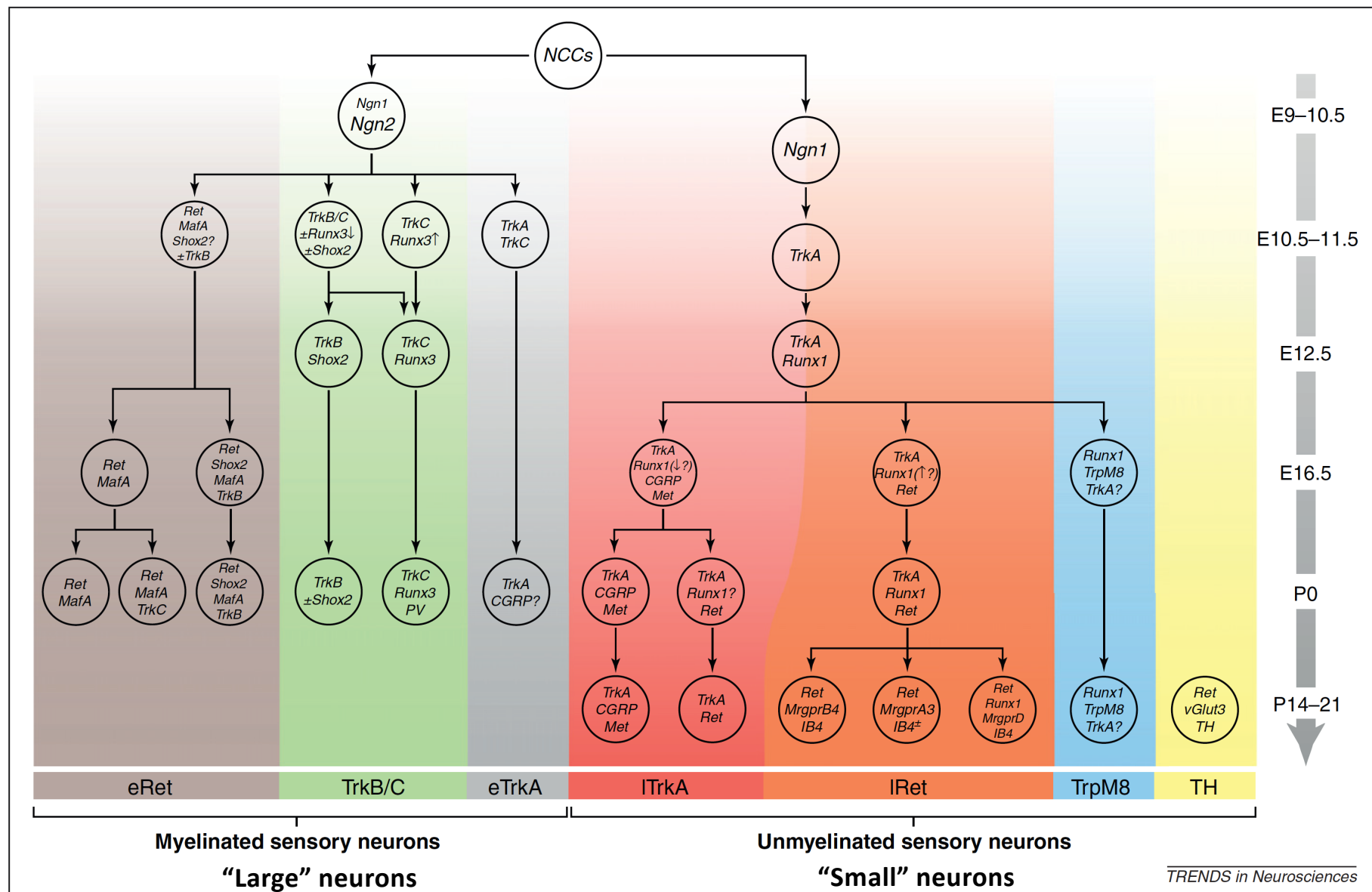
Respond to **different stimuli**:

- Respond to different stimuli:
- Chemical (temperature, mechanical) } Cutaneous receptors
 - Chemicals → Gustation/olfaction
 - Electromagnetic → Vision
- Primary Sensory Nerve Terminals
-
- Receptors and their ligands:
- PAR2: Trypsin, Trytase
 - TRPV4: Pressure, Osmolarity
 - TRPA1: ROS, PGE₁, PGE₂, PGD₂
 - TRPV1: H⁺ (pH ≤ 6.4), Capsaicin, Heat, Anandamide, Leukotriene B4, NGF?
 - P₂X₂: ATP
 - CCKBR: CCK
 - 5-HT_{1A, 1D, 2A, 3A}: 5HT
 - TRK A: NGF
 - HR: Histamine
 - BR: Bradykinin
- Neurotransmitters: CGRP, SP
- Effects: Edema, Inflammatory cells, Necrosis

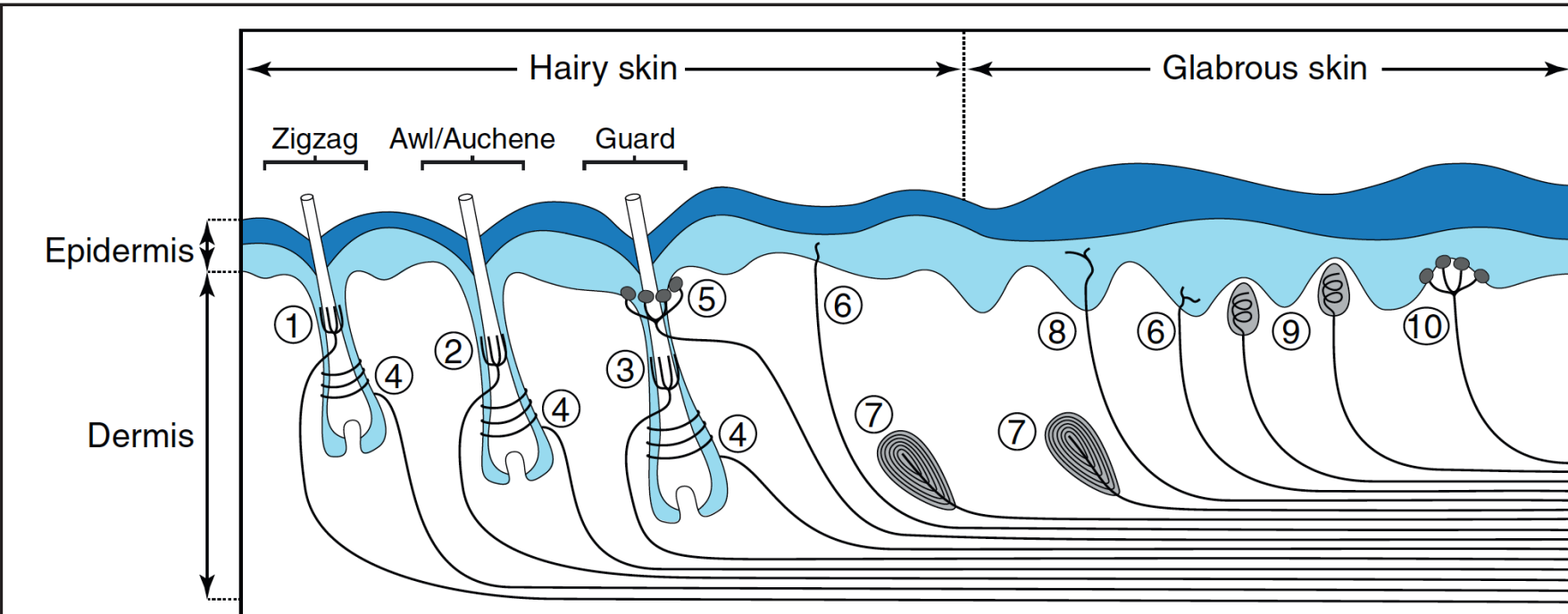
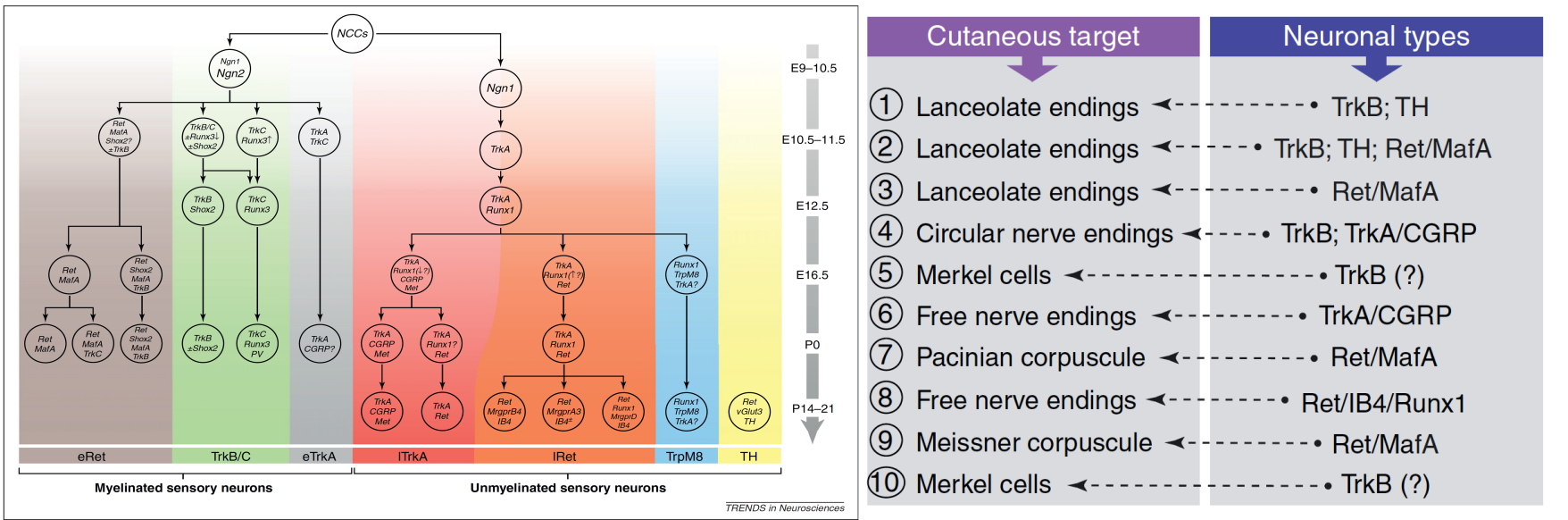
2021 Nobel price in Physiology/Medicine



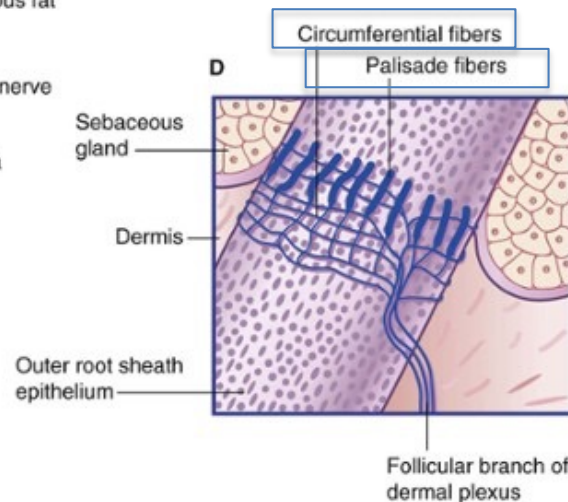
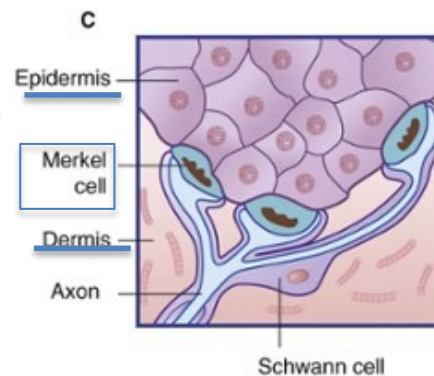
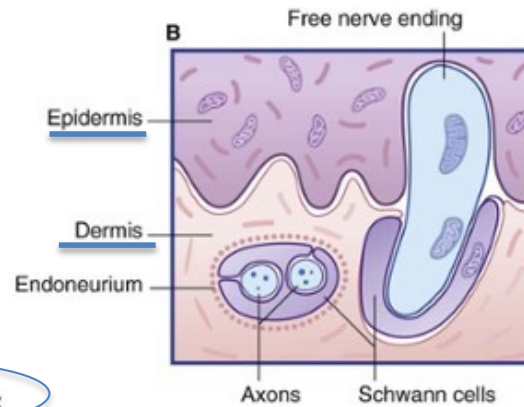
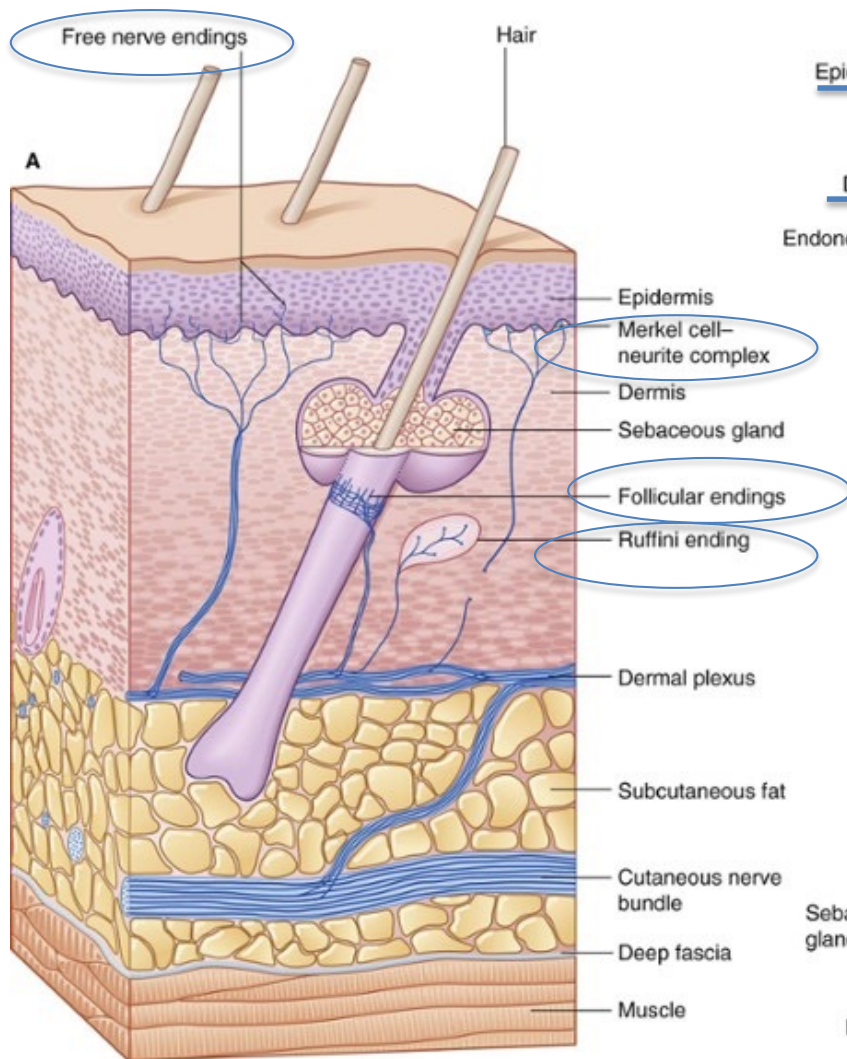
Complex diversification of sensory neurons



Cutaneous sensory system



Non-glabrous (hairy) skin



Free nerve endings (also glabrous)

- **Epidermal** nerve endings
- **Dermal** nerve endings
- Different stimuli:
- **Thermoreceptors**
- **Nociceptors**

Merkel cells (also in glabrous)

- **Cells are forming a complex with the neurite**
- **Sense pressure** for example from edges of objects
- **Slowly adapting**

Follicular nerve endings

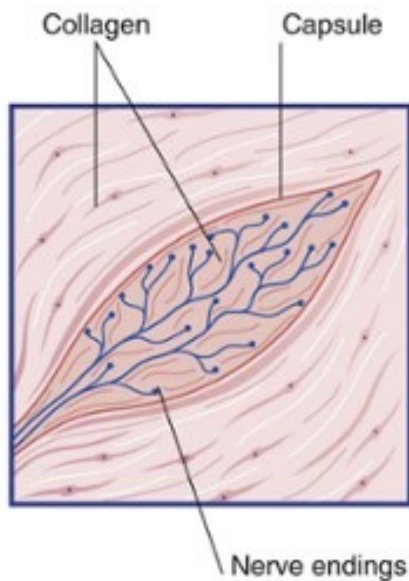
- **Palisade (lanceolate) of naked** nerve terminals
- **Circumferential** terminals
- Each follicular unit **innervate multiple follicles**
- **Rapidly adapting**

Ruffini endings (also glabrous)

Encapsulated nerve endings

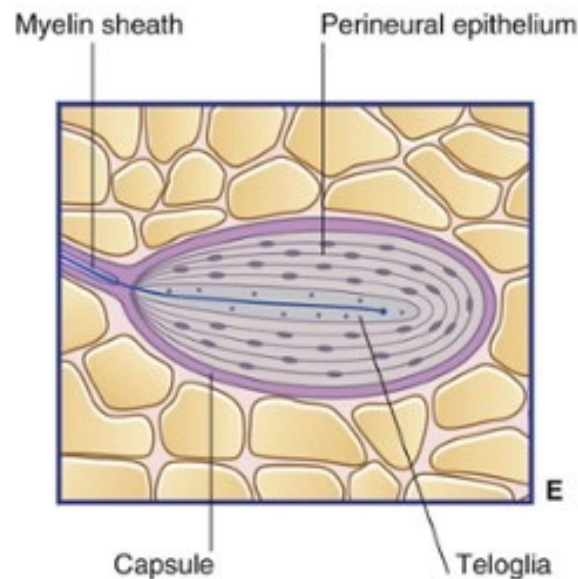
Three types of cutaneous **mechanoreceptor** nerve endings are encapsulated by **capsules** formed by **3 layers: Outer coat of connective tissue, Middle coat of perineural epithelium** and an **Inner coat of modified Schwann cells (teleglia)**

Nerve endings of Ruffini



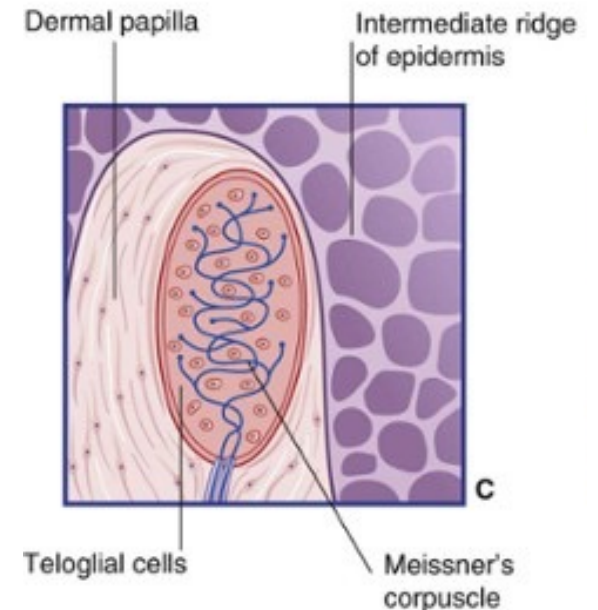
- **Axons branch** liberally in a **collagenous core**
- Respond to **shearing stress**
- In **glabrous and non-glabrous skin**

Pacinian corpuscles



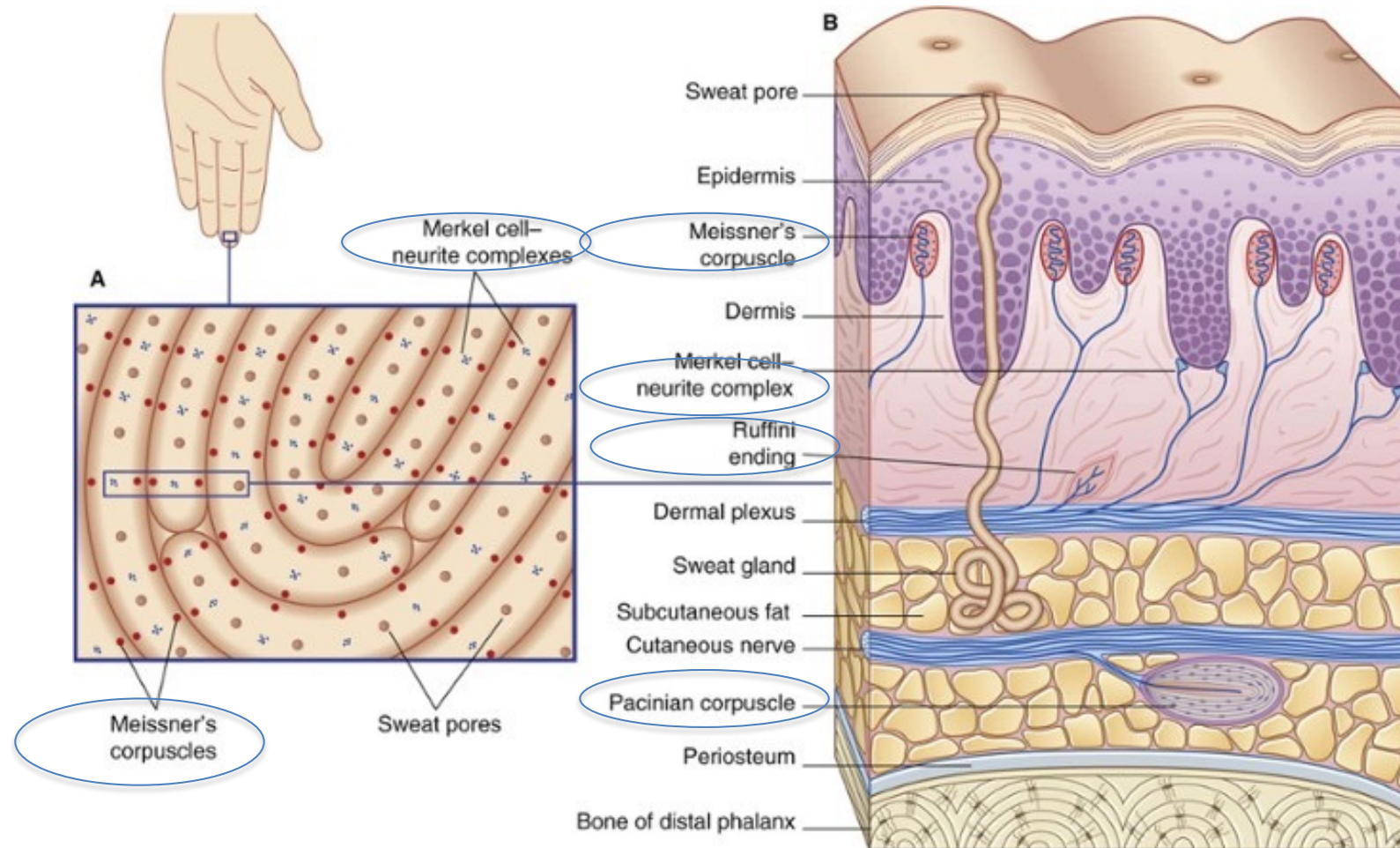
- **Onion-like** structure with **several teleoglial lamellae** surrounding a **single axon**
- Size of a rice grain, ~ 300 in a hand
- **Sub-cutaneous, discharge both when compressed and released** → fire when an object is picked up and again when released
- Only in **Glabrous skin**

Meissner's corpuscles



- **Ovoid-shaped** contain several **axons zig zag among teleoglial lamellae**
- Only in **Glabrous skin**

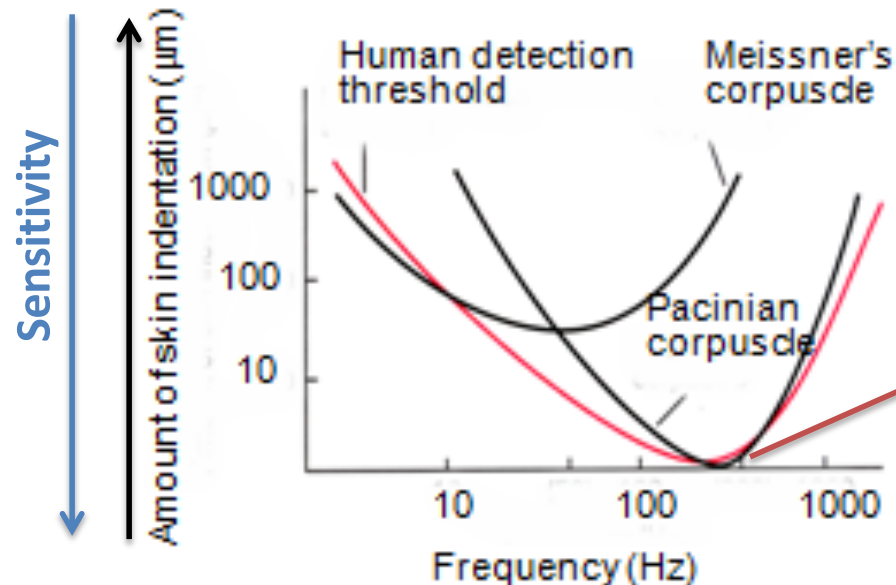
Glabrous skin



- Beside **free nerve endings**, glabrous skin is rich in **Pacini corpuscles**, **Ruffini endings**, **Meissner's corpuscle** and **Merkel cell-neurite complexes**
- **Meissner's corpuscle and Merkel cells** have an **organized distribution on finger pads**
→ together detect fine structures as little as 5 microm in height, such as braille text, fine cloth texture

Vibration Sense

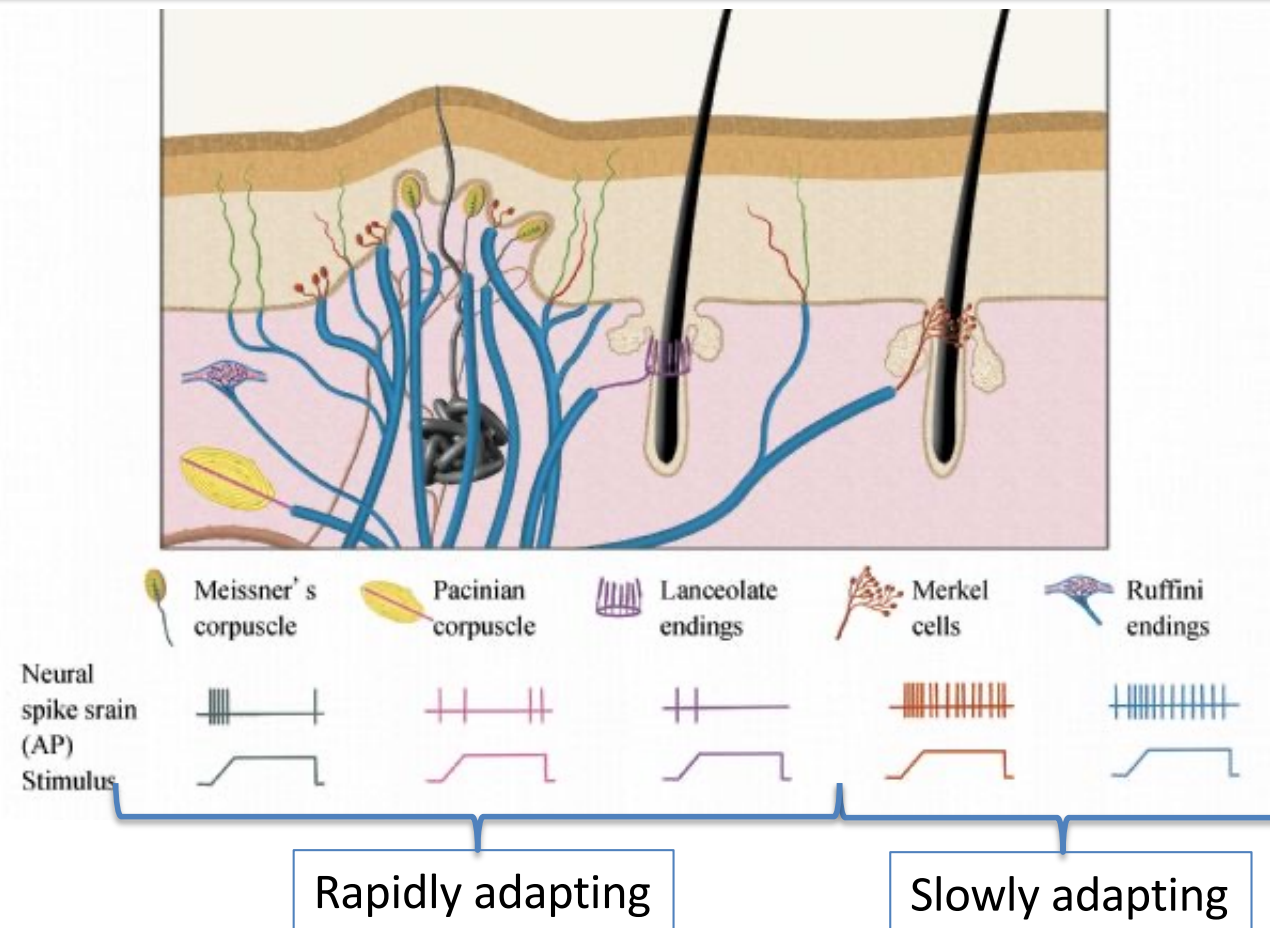
- Vibration sense is detected by rapidly adapting **Meissner's and Pacinian corpuscles (mechanoreceptors)**, each of which has a **different threshold sensitivity to different frequency vibration (in Hz)**:
 - **Meissner's corpuscles**: 5-300Hz; most sensitive at 20-50Hz
 - **Pacinian corpuscles**: 60-400Hz; peak at 250 Hz
- These **sensitivity overlap**, so for **middle frequency** vibrations, they will be **firing simultaneously**. The receptors generate an action potential for every 'cycle' of vibration.



Due to Pacinian corpuscles that are highly sensitive at 250 Hz → **body is most sensitive to vibrations of around 250Hz.**

Threshold for detecting vibration corresponds to the tuning threshold of the mechanoreceptor

Cutaneous receptors have different adaptation



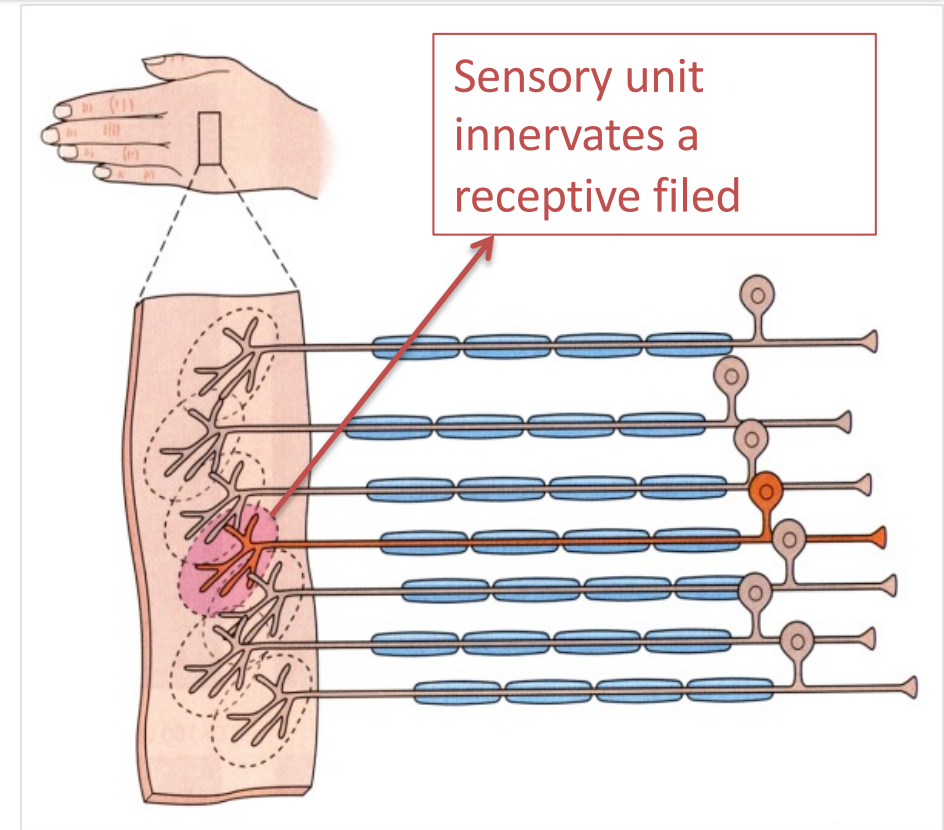
- Lanceolate **follicular endings** are **rapidly adapting** → rapidly unaware of clothes
- **Pacinian corpuscle** and **Meissner's corpuscles** are **rapidly adapting**
- **Merkel cells** and **Ruffini ending** are **slowly adapting** → continue to fire when holding an object

Acuity and two-point discrimination of touch

The **acuity and two-point discrimination of touch sensation** differs at different areas of the body.

This is mainly due to two factors:

- the **concentration of sensory receptors in that region** : es. the more that are present, the more likely more than one will be innervated by the same stimulus → increased acuity
- the **size of the receptive fields** of the specific sensory unit, **which depends on receptor type**: es. the smaller the receptive fields, the more accurate a **representation of the stimulus** is signaled to the brain → higher two-point discrimination



A
Receptive field of Pacinian corpuscles



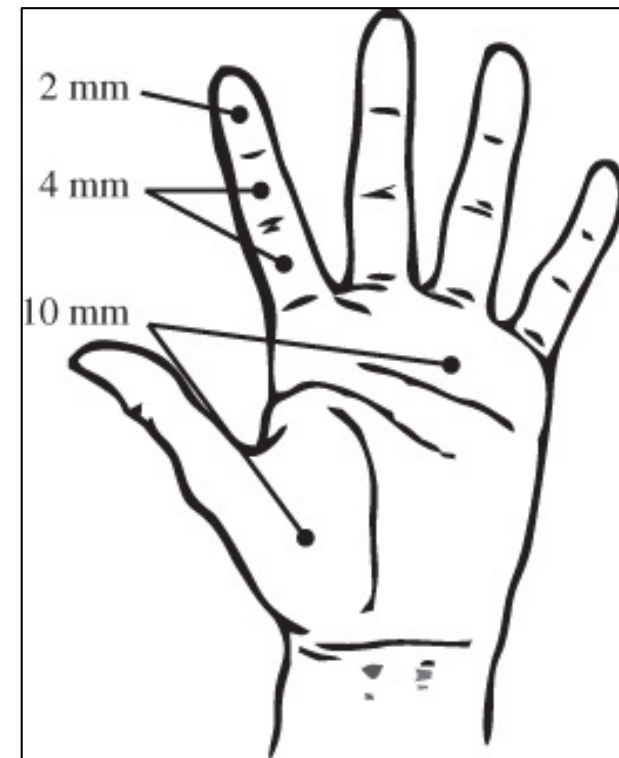
B
Receptive field of Meissner's corpuscles



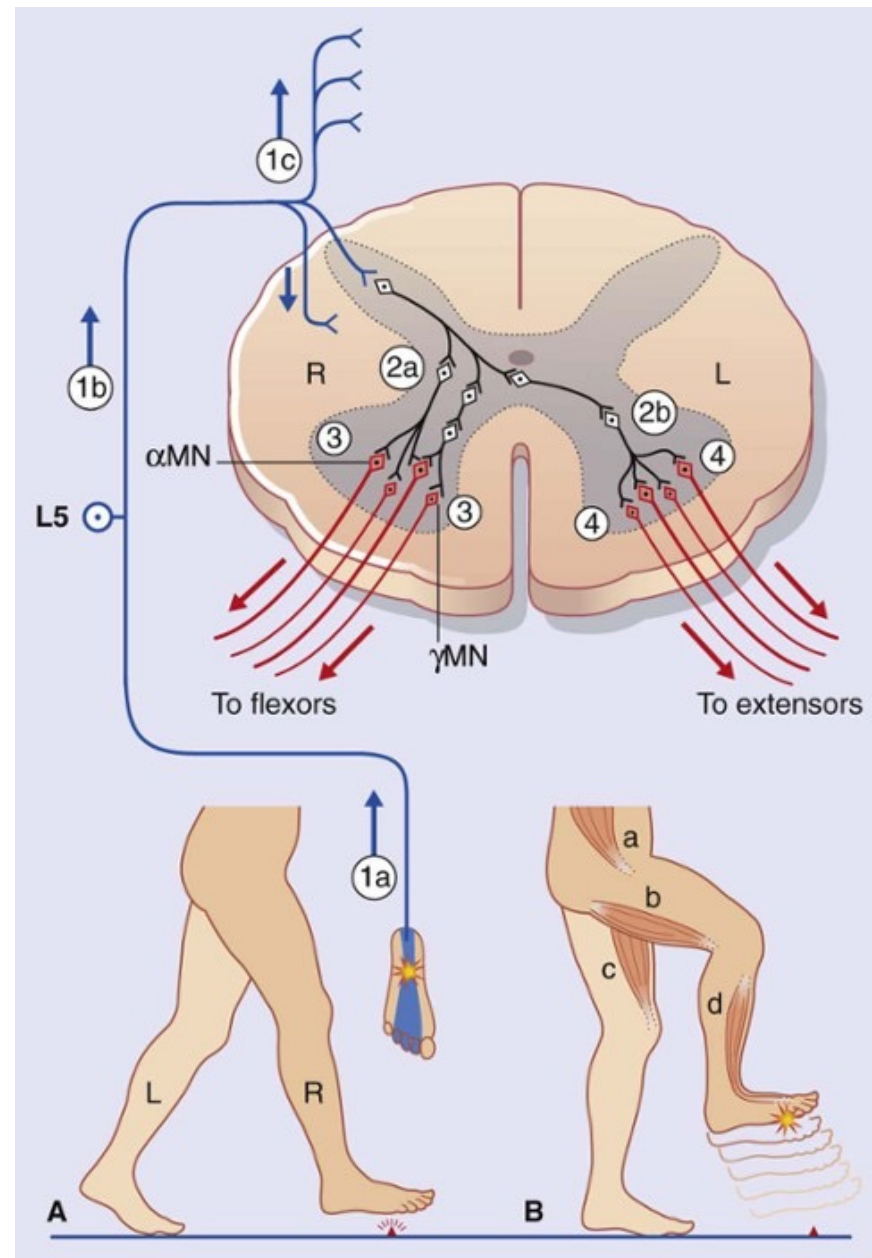
Lateral inhibition

- **Two-point discrimination is improved by lateral inhibition.** This is the process by which **neurons inhibit signals from nearby neurons** via an intermediate neuron in the CNS. This **enhances the difference between strong signals (at the point of the stimulus) and weaker signals (generated nearby)**, thus the body can pinpoint more exactly where the stimulus is coming from.
- **Different parts of the body are different at two-point discrimination:**

Scheme of discrimination sensation		
	Ending of tongue	1,1MM
	Tips	2,2MM
	Red part of the lips	4,5MM
	The back of tongue	9,0
	The skin of cheek	11,2
	The back of hand	31,5MM
	The forearm	40,5MM
	The leg	40,5MM
	The upper part of trunk	53,0MM
	The middle part of trunk	67,5MM
	The back of neck	67,5MM



Exteroception and Reflexes

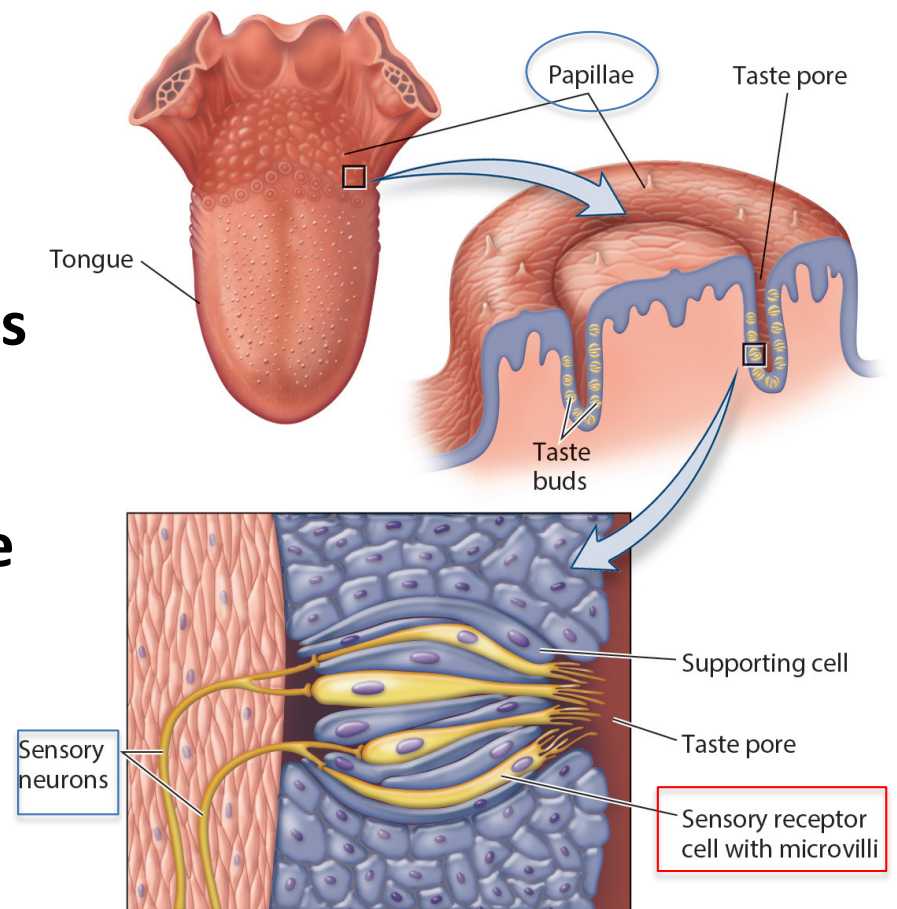


Gustation and Olfaction

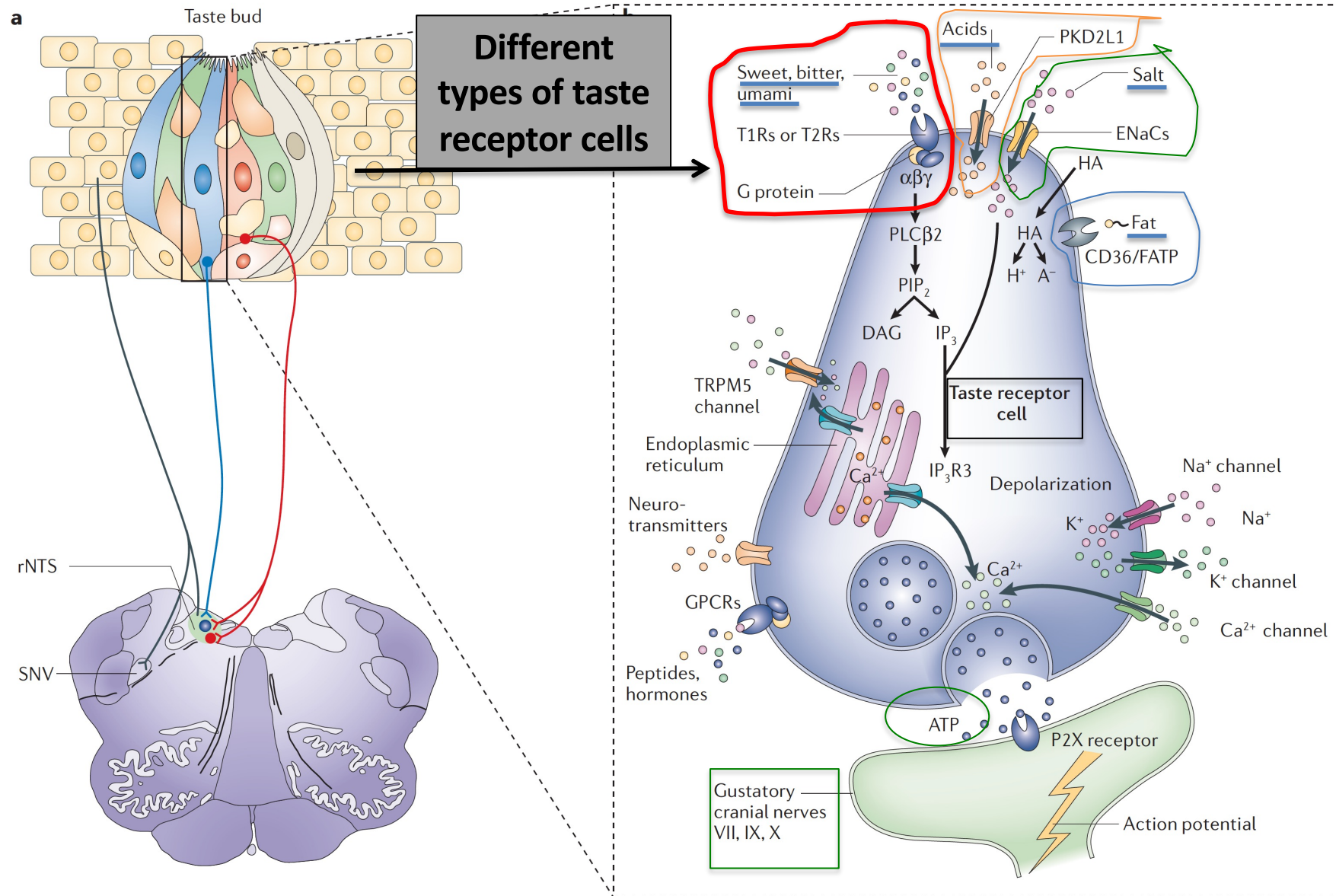
- **Olfaction** – sense of smell
- **Gustation** – sense of taste
- Chemicals in air, water, or food bind to **chemoreceptors**, which initiate electrical responses in other neurons that pass into the brain
- Close relationship between taste and smell
 - Distinction is meaningless for **aquatic animals**

Gustation

- Many animals use sense of taste to select appropriate foods
- **Taste buds** are chemosensory cells that detect particular **molecules in food**
- **Papillae collect food molecules** and **direct** them through the taste pore **to receptor cells in taste buds**
- Tips of **sensory receptor cells** have **microvilli** that extend **into taste pore**



Sensory receptor cells

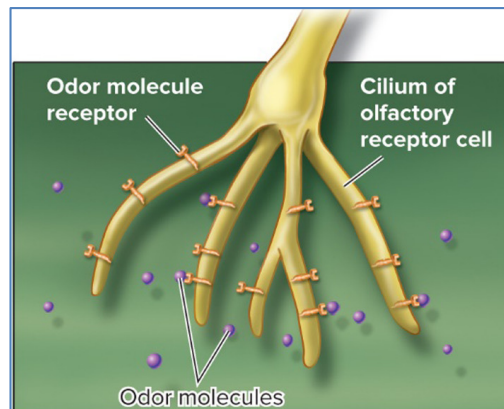
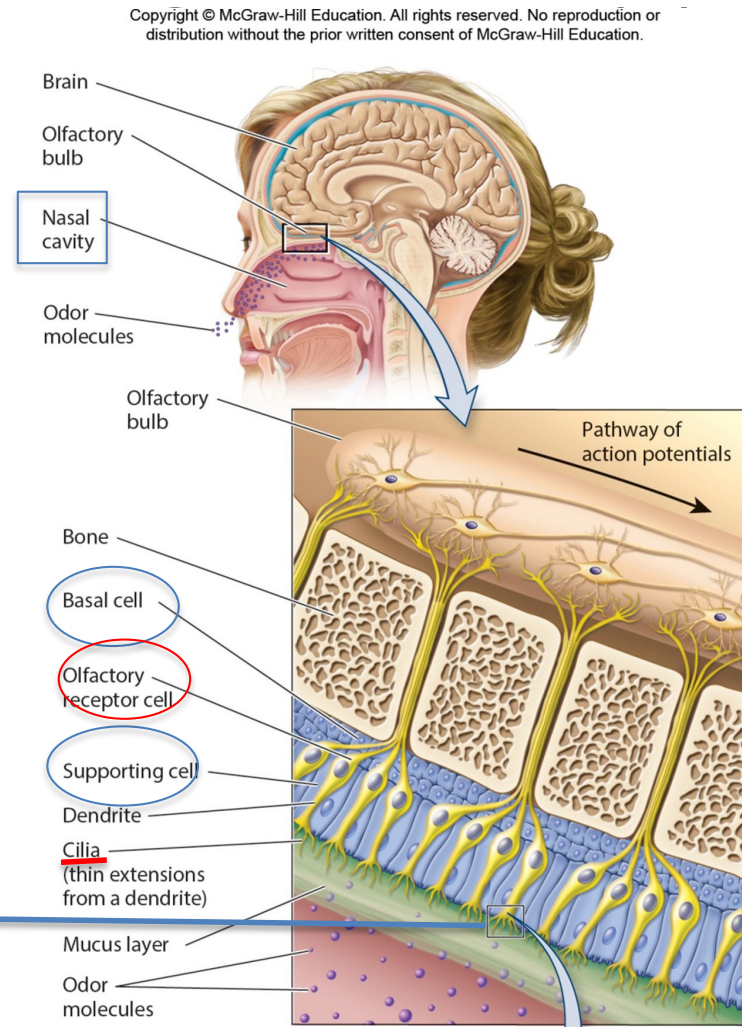


Olfaction

Olfactory sensitivity depends on **number of olfactory receptor cells**

Only 5 to 6 million in humans versus 220 million in dogs

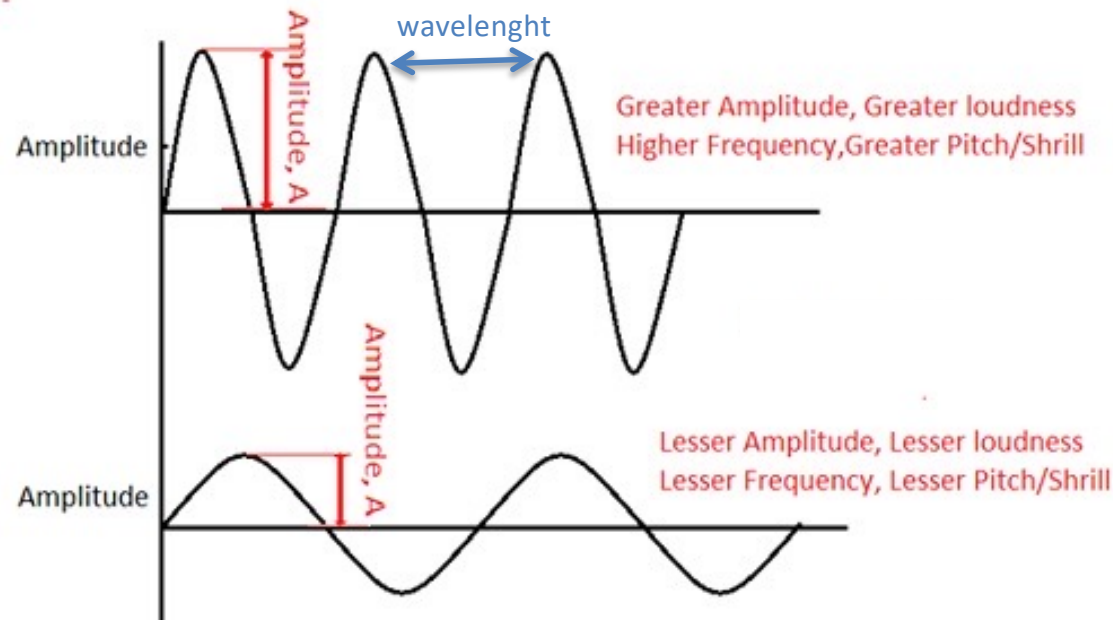
- **Olfactory sensory receptors – neurons** located in the epithelial tissue at **upper part of nasal cavity**
- **Surrounded** by two cell types:
 - **Supporting cells** provide physical support
 - **Basal cells** differentiate into new olfactory receptor cells



a) Olfactory receptor cells in olfactory epithelium

Audition

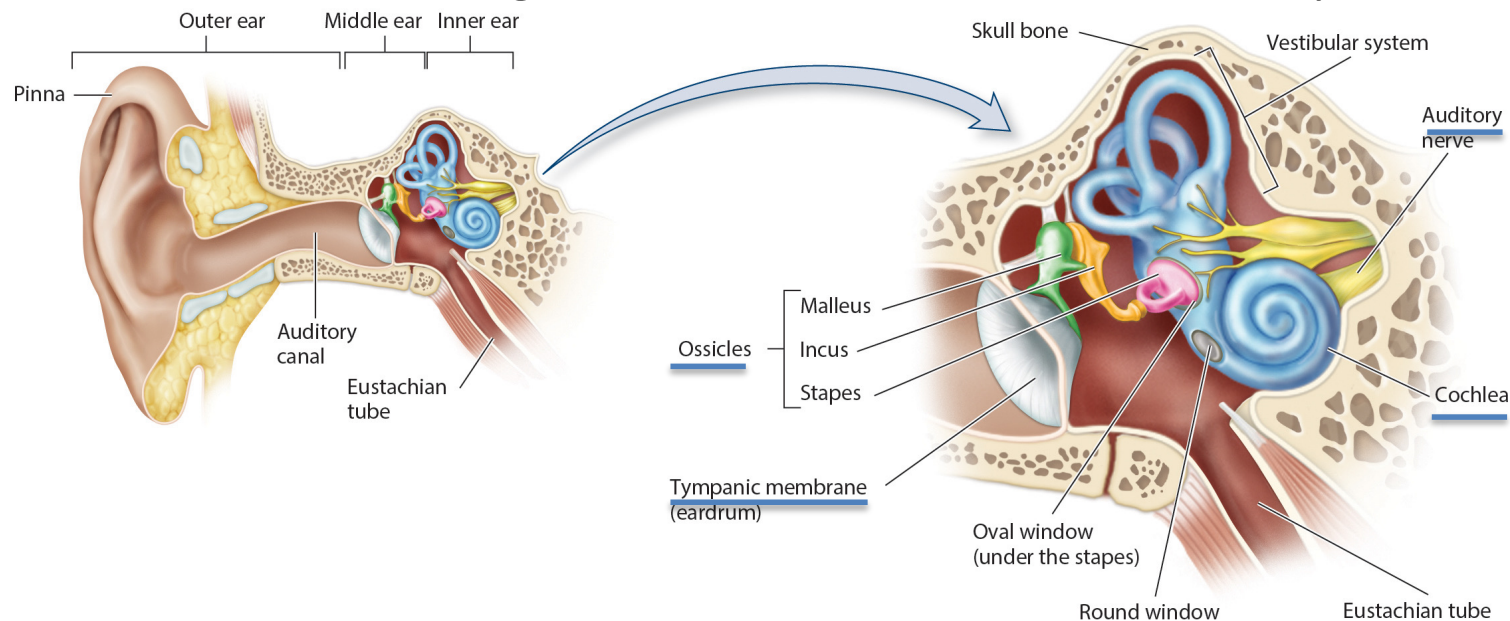
- Ability to detect and interpret sound waves (speed ~ 330 m/s in air)
- **Wavelength** – distance from the peak of one wave to the next
- **Frequency** – number of complete wavelengths in a second (Hertz)
- **Short wavelengths** have **high frequencies** perceived as **high pitch** or tone
- **Long wavelengths** have lower frequencies perceived as a **lower pitch**



- **Amplitude** $\rightarrow$ **loudness**

Mammalian ear

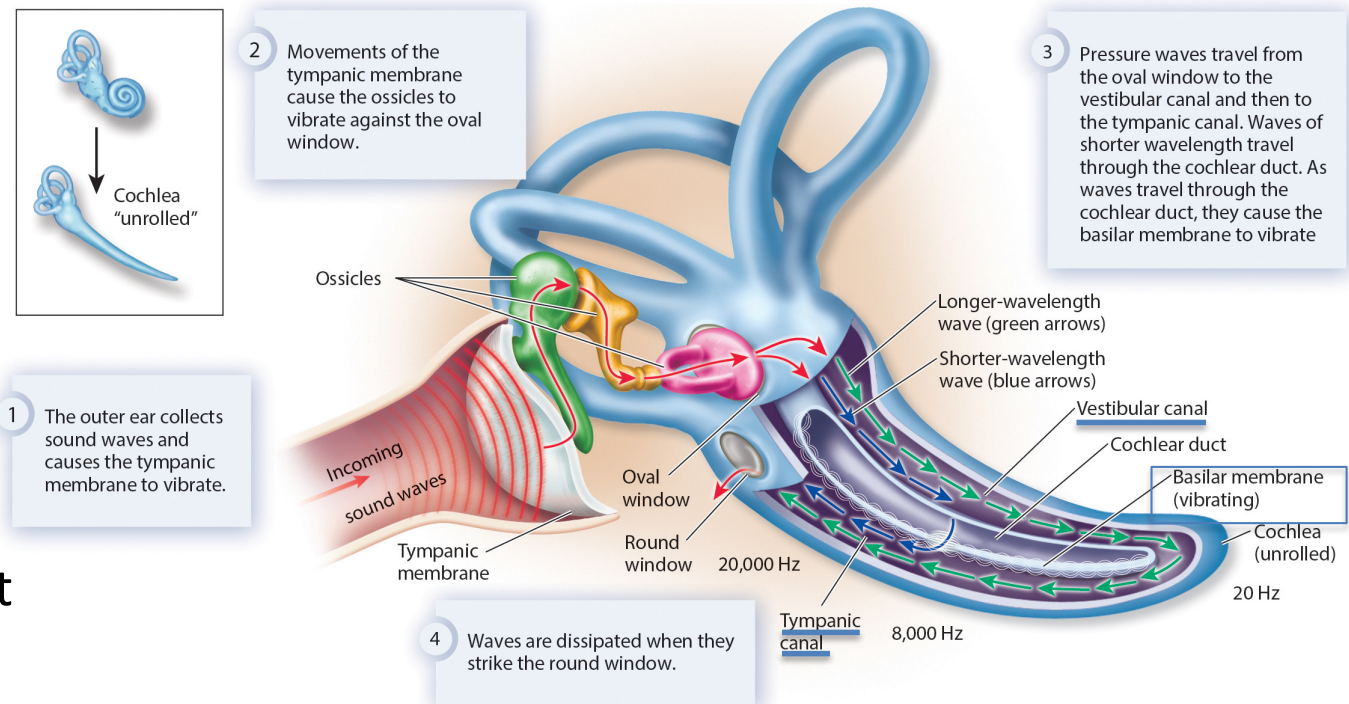
- Three compartments:
 - **Outer ear – pinna and auditory canal**
 - Separated from middle ear by tympanic membrane (eardrum)
 - **Middle ear – ossicles** (malleus, incus, and stapes) connect eardrum to oval window
 - Separated from inner ear by oval window
 - **Inner ear – cochlea and vestibular system**
 - Generates signals that travel to brain via auditory nerve



Transduction of mechanical vibrations

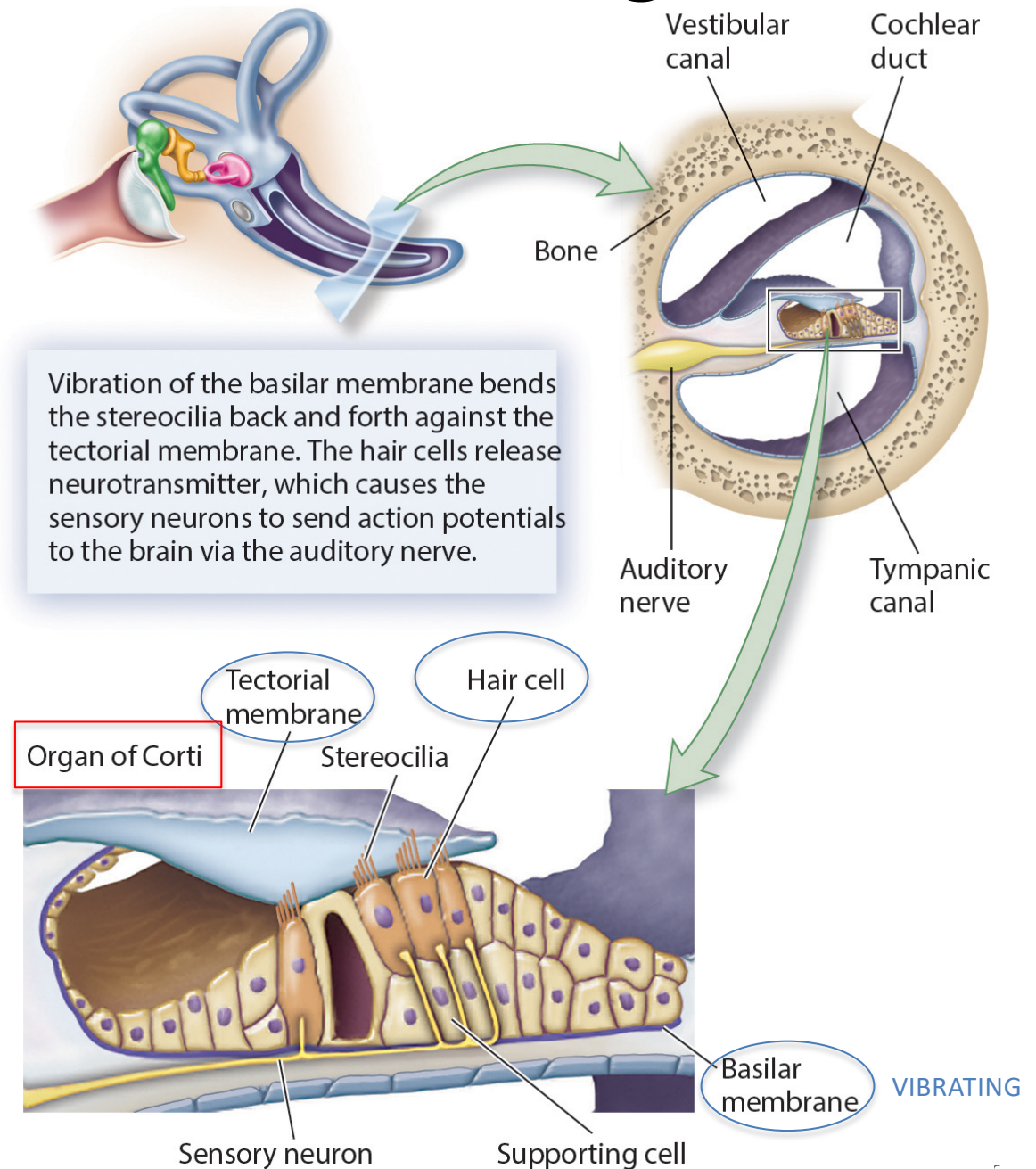
- Sound waves enter the **outer ear** and **Tympanic membrane** vibrates back and forth
- **Ossicles** transfer vibration to **oval window**, causing it to vibrate against the cochlea → sends **pressure waves** through **perilymph**
- **Waves travel** from **vestibular canal** to **tympanic canal** and dissipate against **round window**:
 - Low frequency sounds – complete route
 - Higher frequency sounds – pass through cochlear duct
- Cause vibration of **basilar membrane**

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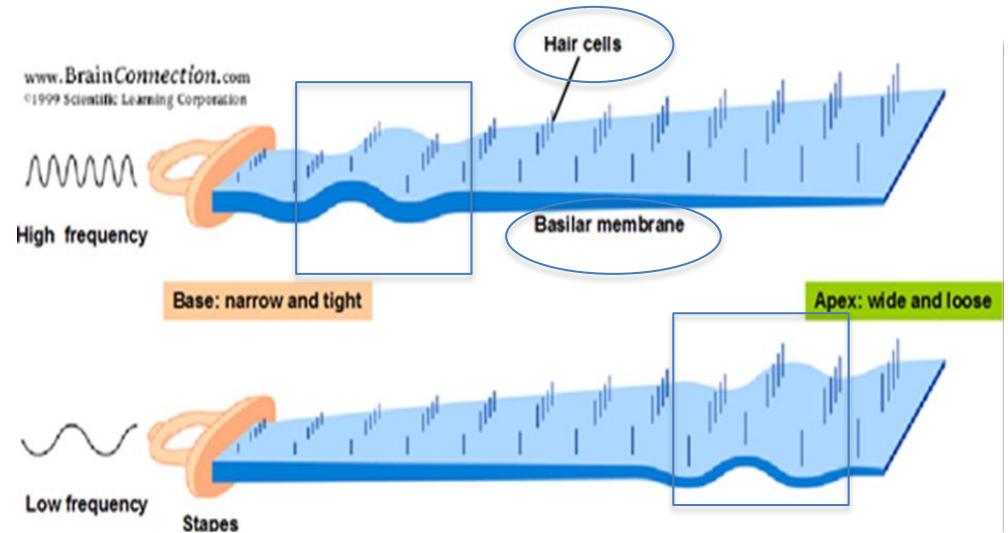
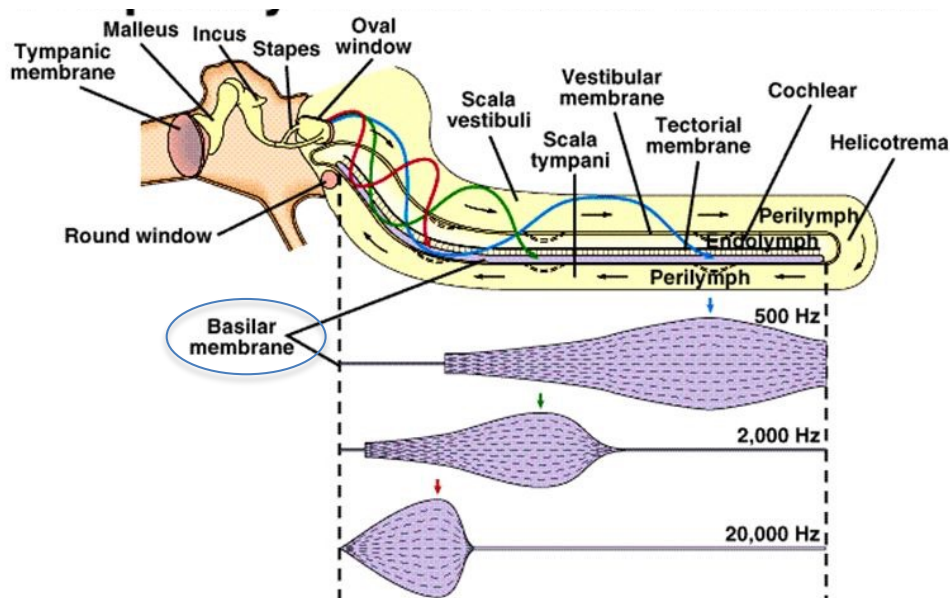


Generation of electrical signals

- Mechanical vibrations are transduced into electrical signals within the **organ of Corti** of the cochlea
- Hairs bending in one direction sends an **action potential**
- Frequency of action potentials is determined by the **vibration of the basilar membrane**



Loudness and pitch discrimination



The basilar membrane is :

- More rigid at the **base of the cochlea**:
→ **High frequency waves** vibrate them
- Less rigid **near the end**: → **Low frequency waves** vibrate them

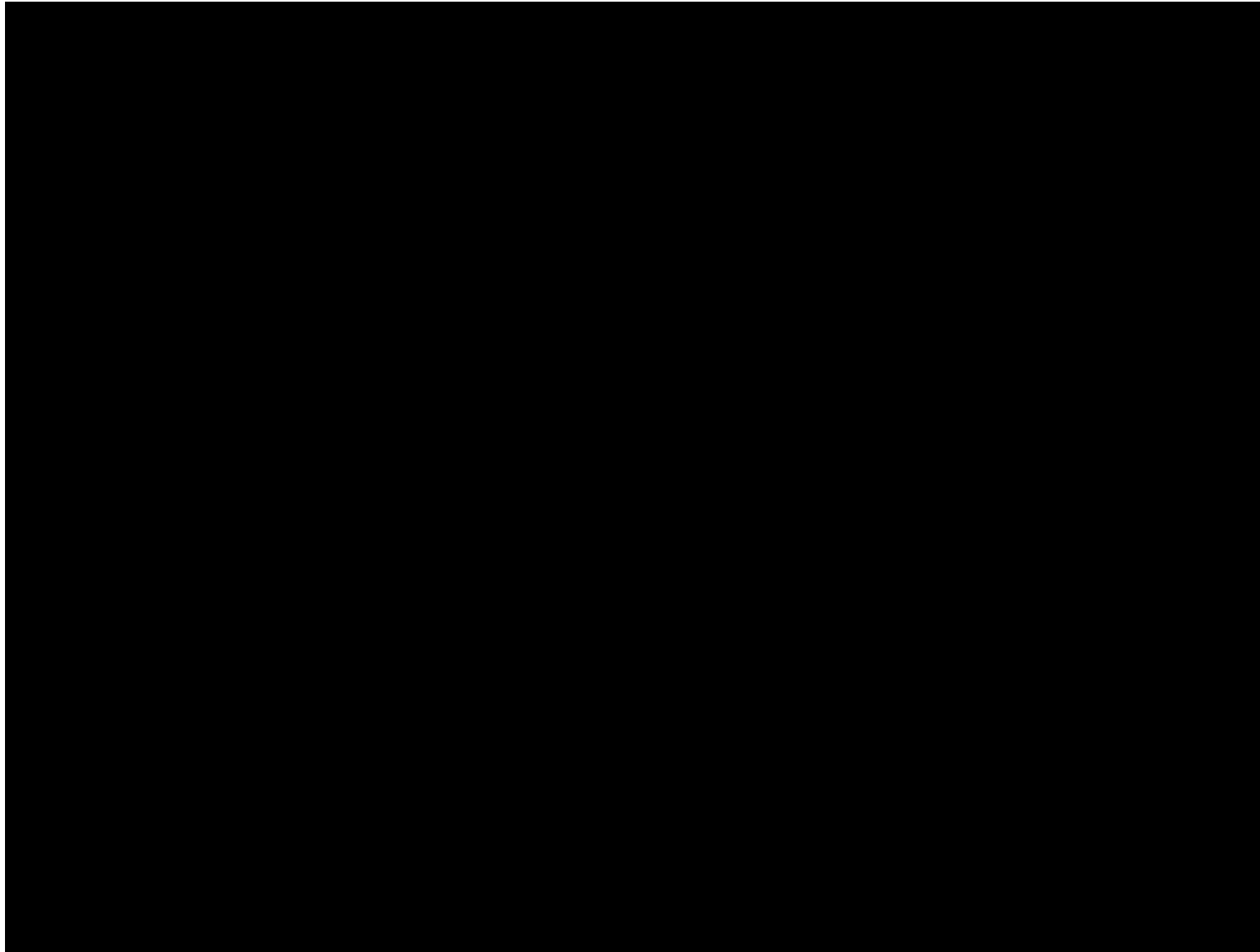
Intensity discrimination

Greater the degree of displacement of the hair cells
more hair cells displaced
hence more nerve fibres are stimulated

Transduction of mechanical vibrations-1

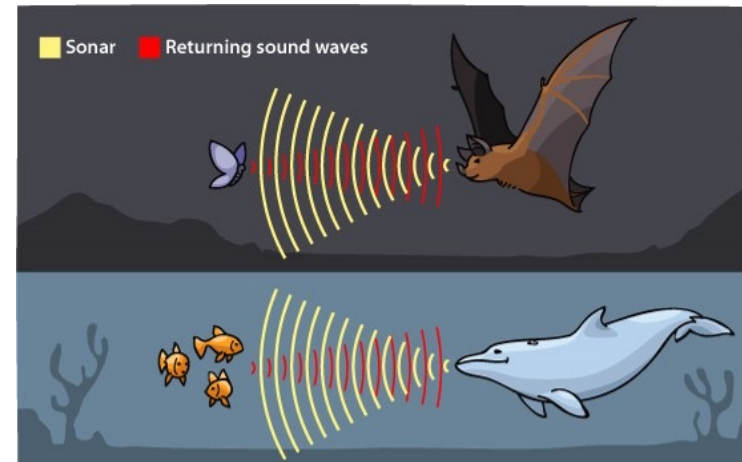


Transduction of mechanical vibrations-2



Hearing in animals

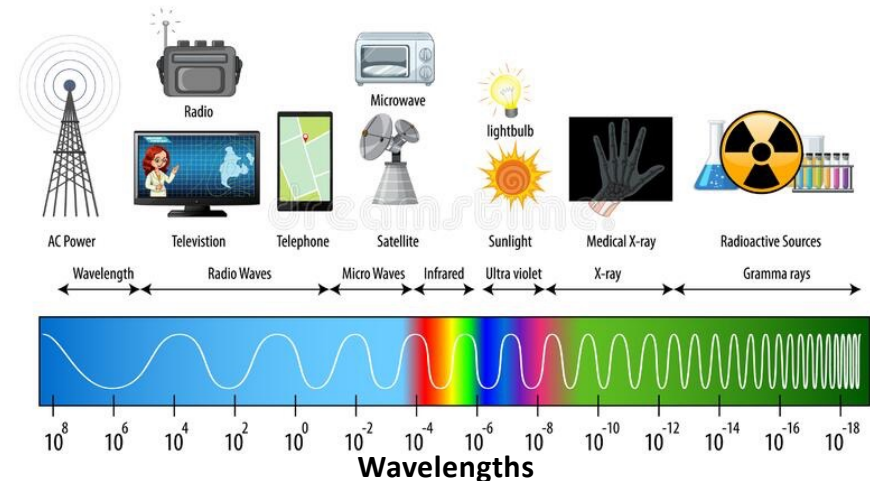
- Humans can hear between **20 to 20,000 Hertz**
- **Bats, toothed whales, and moths** hear highest frequencies
- Baleen **whales** and **elephants** hear lower frequencies
- Many animals rely on **hearing to locate** prey, mates, etc.
 - Sound usually **does not arrive at both ears simultaneously**
 - **Brain can interpret time difference** to determine direction and **origin of sound**
 - Some animals (for example, bats and whales) use **echolocation** – generation of **high-frequency** sound waves to determine an object's location



Electromagnetic reception

- **Detect radiation** within a **wide range of the electromagnetic spectrum**: → Includes wavelengths that correspond to **visible light, ultraviolet light, and infrared light**, as well as **electrical and magnetic stimuli**

THE ELECTROMAGNETIC SPECTRUM



- **Pit vipers** sense heat from prey as infrared radiation



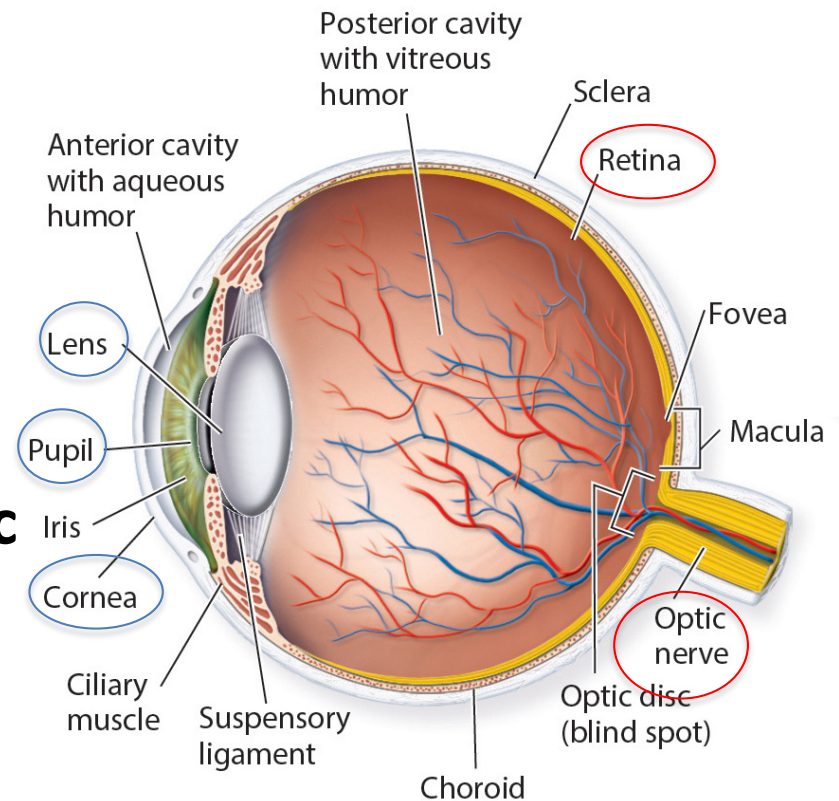
VISION: Photoreception

- **Specialized** form of **electromagnetic reception**
- Photoreceptors detect **photons of light** arriving from the sun or other light source, or reflecting off an object
- **Photon** – **fundamental unit of electromagnetic radiation** with the properties of both a particle and a wave
- **Different vision organs** in different organisms
 - Most invertebrates → Eyescups (for example in planaria) or **compound eyes** (in insects)
 - **Vertebrates** and some invertebrates (squid, octopus, and in some snails and annelids) → **single-lens eyes**



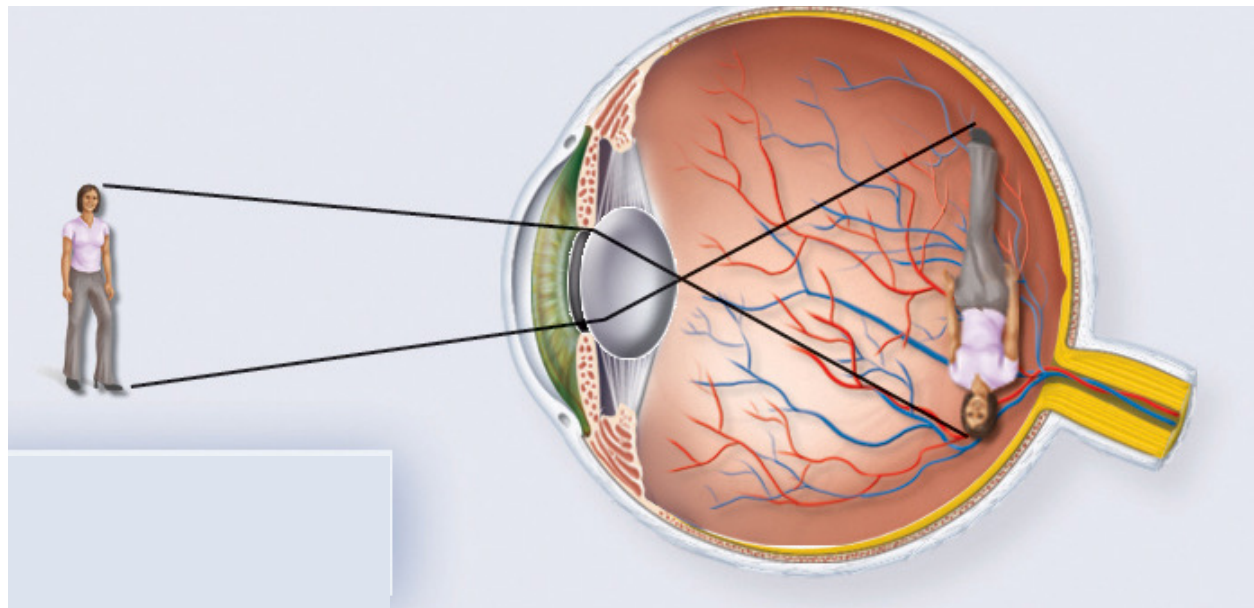
The single-lens eye

- **Light transmitted through cornea, pupil, lens, to layer of photoreceptors (the retina) at the back of the eye**
 - Visual image is formed on the retina
- Photoreceptors trigger **electrical changes in neurons**
 - Pass out of the eye through the **optic nerves** to brain



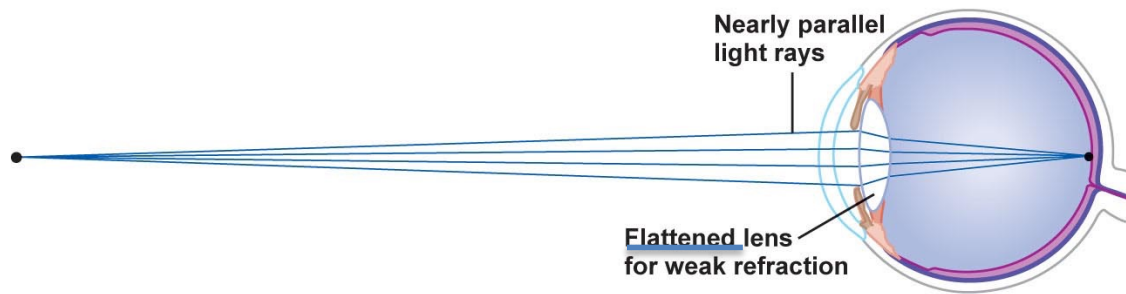
Light focusing

- **Cornea and lens refract light inward** toward photoreceptors at the back of the eye
 - Results in an **upside-down and laterally inverted** image **on the retina**
 - Brain perceives image correctly**

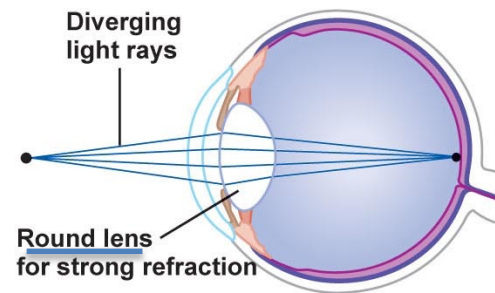


Light focusing: accommodation

- Lens adjusts to focus light coming from different distances
 - In fishes and amphibians, moves forward or backward
 - In birds and **mammals**, **lens** remains stationary **but changes shape**



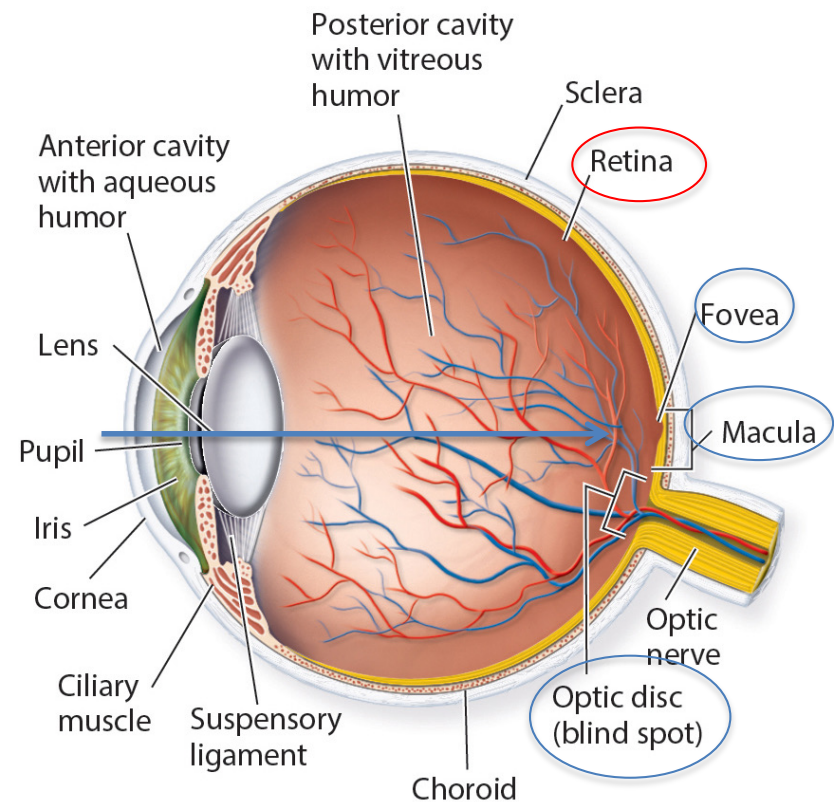
(a) Viewing a distant object



(b) Viewing a near object

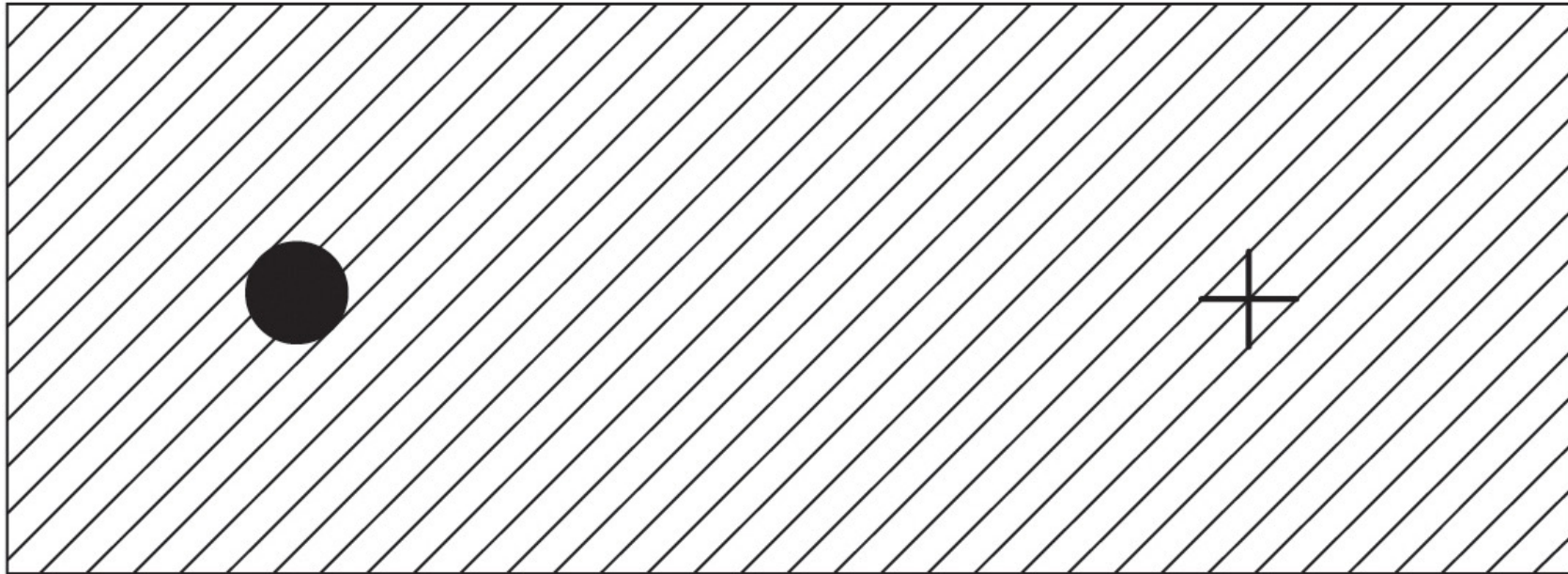
The retina

- **Macula – region directly in line with pupil and lens**
 - **Fovea** – located in the center of macula
 - **Highest density of photoreceptors for color**
 - **Responsible for sharpness of daylight vision**
- **Optic disc (macula cieca)**
 - **Point on retina where optic nerve exits**
 - **No photoreceptors so no image on that area → Blind spot**



Demonstration of blind spot

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Chiudete occhio sinistro e fissate con occhio destro il punto. **Spostandovi avanti ed indietro** ad un certo punto non si vedra' piu' la croce perche' la luce riflessa dalla croce cadra' appunto nel blind spot!

Rods and cones

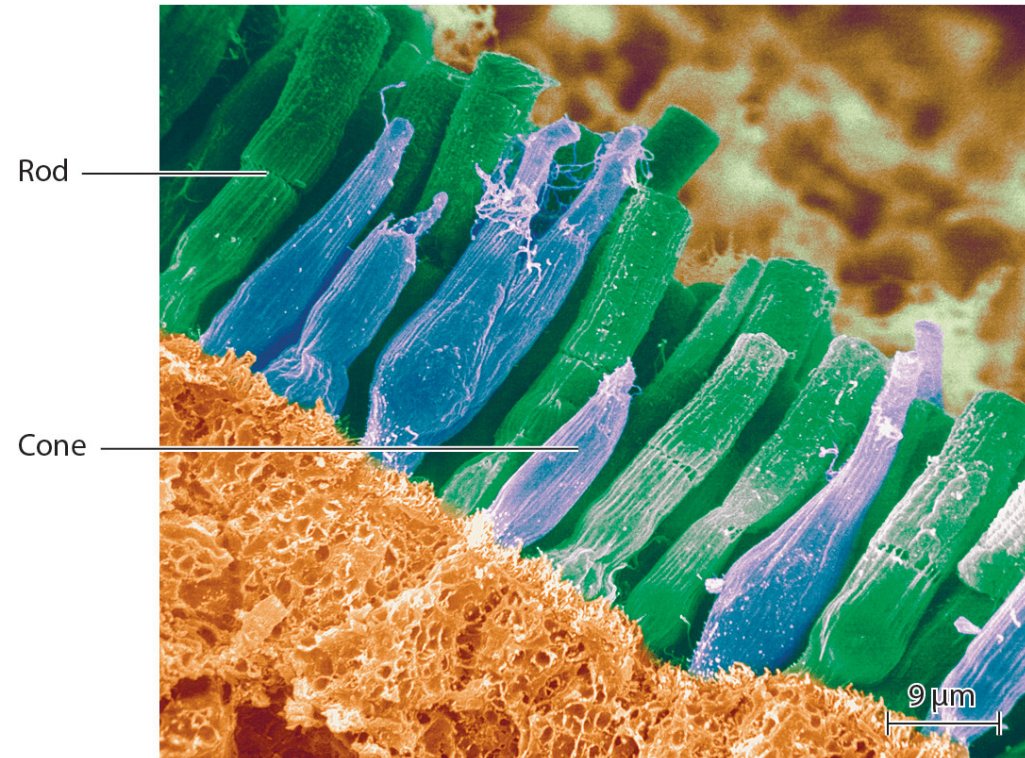
- **Rods**

- Sensitive to **low-intensity light** (up to a single photon)
- Used mostly **at night**
- Do **not detect colors**
- Generate a **black-and-white** image
- Nocturnal animals rely primarily on rod vision

- **Cones**

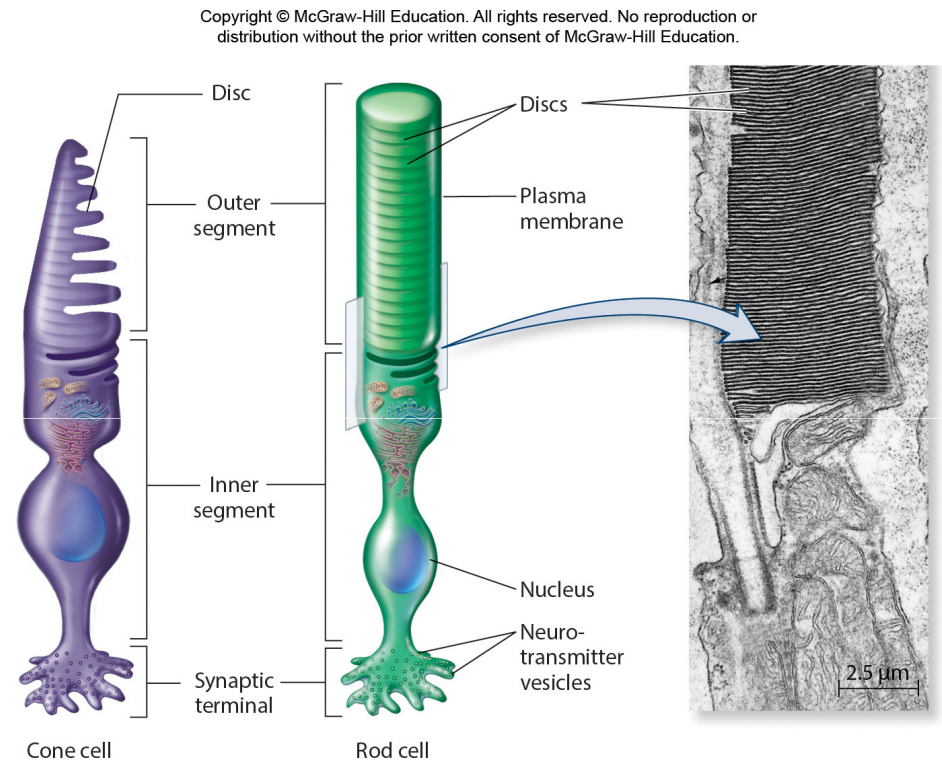
- Require **more light**
- Used in daylight by most **diurnal** vertebrates and some insects
- **Detect color**
- **Fewer cones than rods** in human retina

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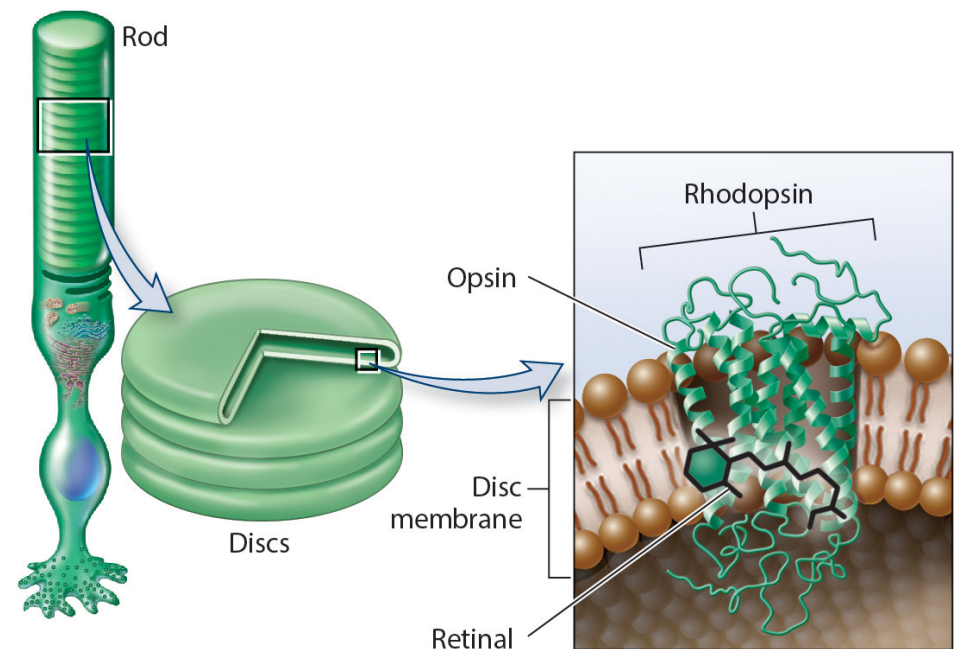
Three functional parts of rods and cones

- **Outer segment**
 - **Contains folds of membranes** that form stacks like discs
 - Discs contain **pigment** that **absorbs light**
- **Inner segment**
 - Contains nucleus and other organelles
 - Synaptic terminal with **neurotransmitter**-containing vesicles
- No axons



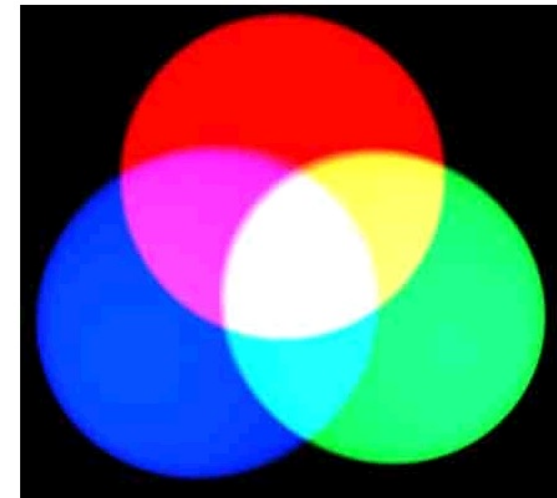
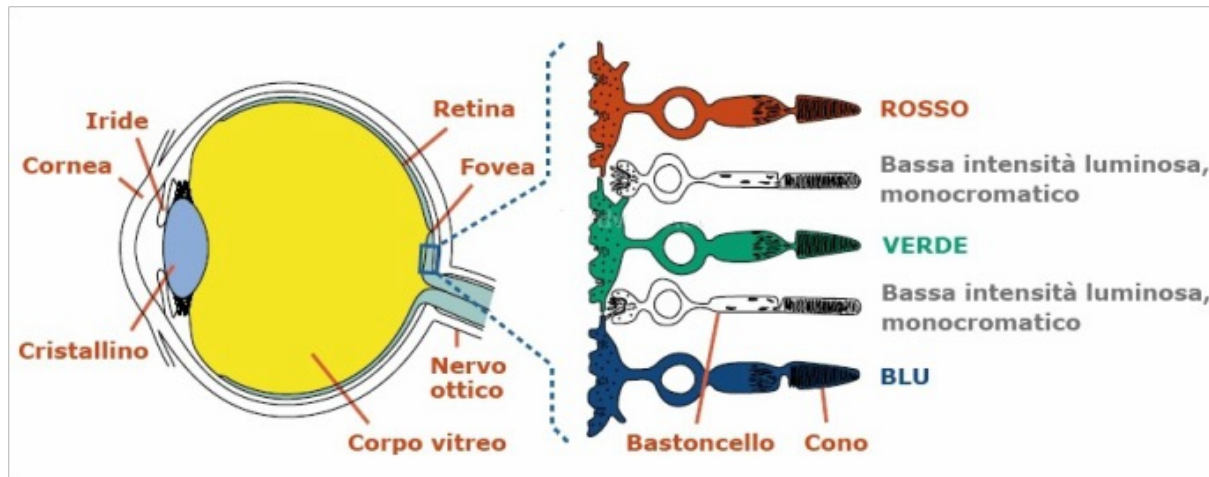
Visual pigments

- Molecules that **absorb light**
 - Embedded in **disc membranes of rods and cones**
- Consist of **two compounds bonded together**
 - Retinal** – **vitamin A derivative**, absorbs light energy
 - Opsin** – **G-protein-coupled receptors** that change membrane **permeability to ions**
- Several types:
 - **Rhodopsin** in rods
 - **Photopsins** (or cone pigments) in cones



Human cone pigments (photopsins)

- Composed of **retinal** plus **1 of 3 possible opsin proteins**
 - Each **type of opsin** determines **wavelength absorbed**
 - Responds best to either **red, green, or blue light**
 - Each cone cell makes only one type



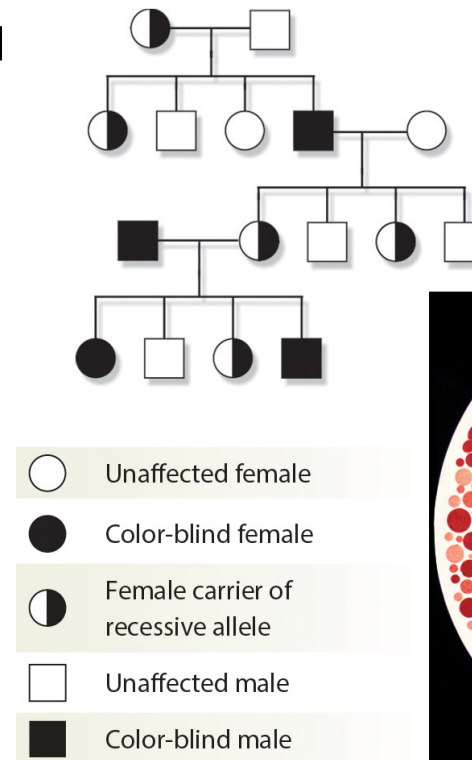
- Brain uses information** about proportion of each type of cone stimulated **to generate all other colors**
- Humans have 3 types of color-sensitive photoreceptors**, whereas **many vertebrates have 4**, some **insects have six types** → **Honeybees can see ultraviolet** wavelengths that humans can't!

Color-blindness

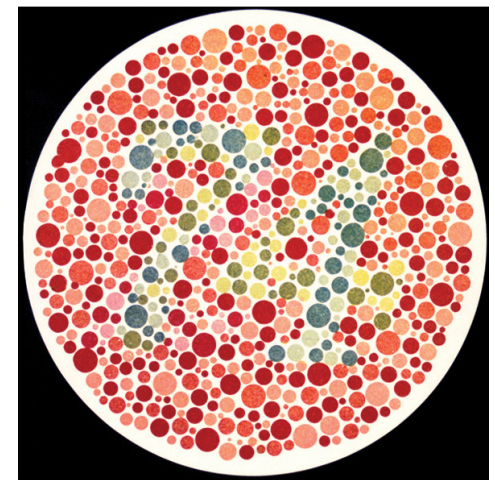
- About **8% of human males** and 1% of human females experience problems in color vision
- **Red-green color blindness** most common
- Lack or have abnormal **red or green cone pigments**
 - In a frequent form **green pigment** responds to **both red and green light**

→ **Poor discrimination between green and red**

- Red and green **opsin genes located on X chromosome** (blue on an autosome)
- In males, the presence of only one X chromosome means that a **single recessive allele** from the mother results in color blindness, even though the mother is not color blind



(a) Pedigree for color blindness



(b) Test for red-green color blindness

Photons alter visual pigments

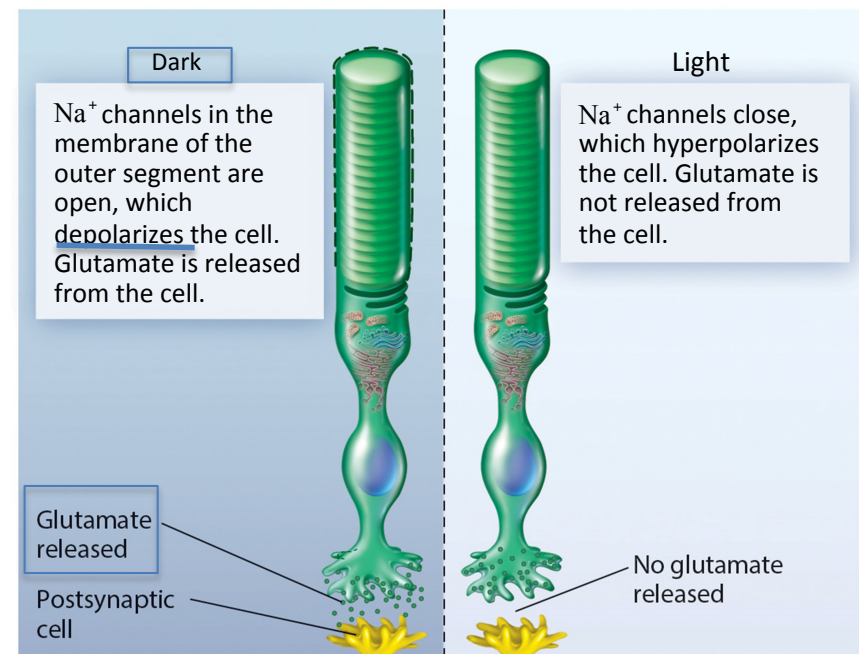
- Photoreceptors differ from other sensory receptor cells because their membrane potential is in a **slightly depolarized at rest**
- **Photoreceptors hyperpolarize rather than depolarize** in response to a stimulus
- Depolarization in unstimulated state results in **continuous release of glutamate**

—Interpreted as absence of light

- **When exposed to light, cell hyperpolarizes**

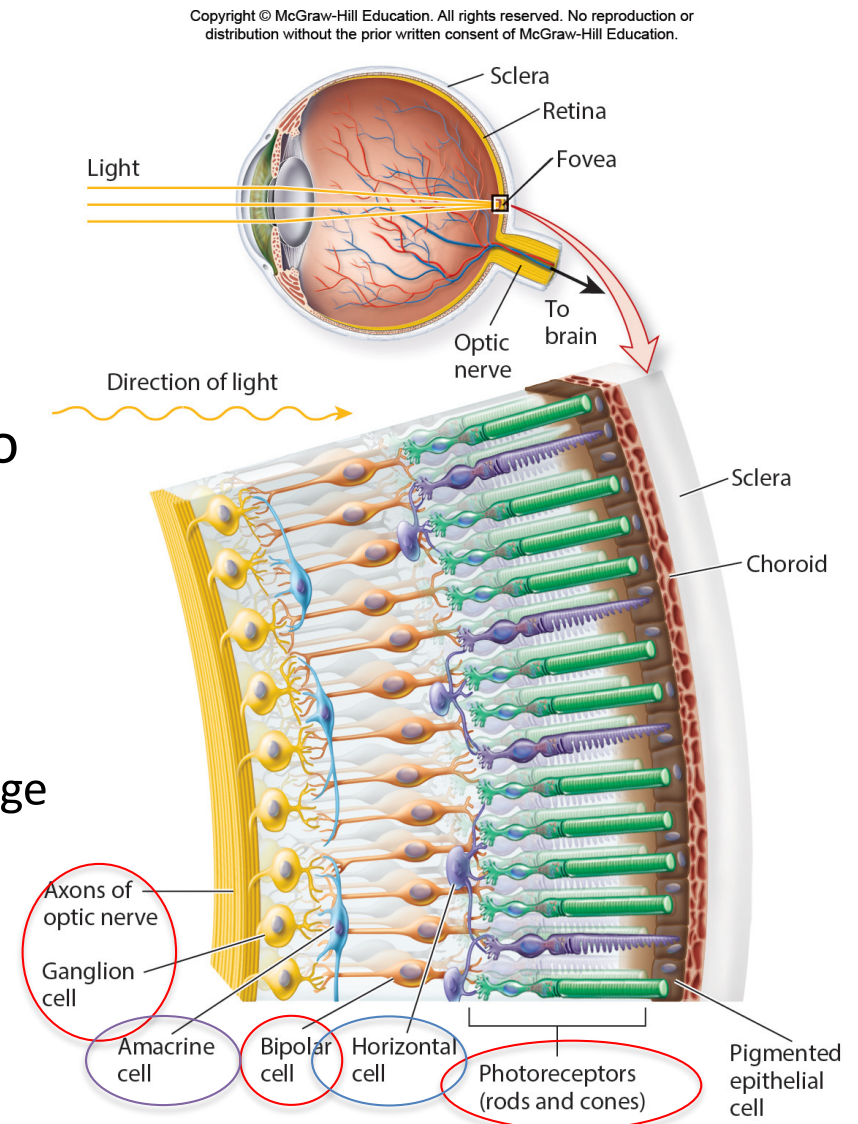
—**Glutamate release stopped**

—Interpreted as visual image



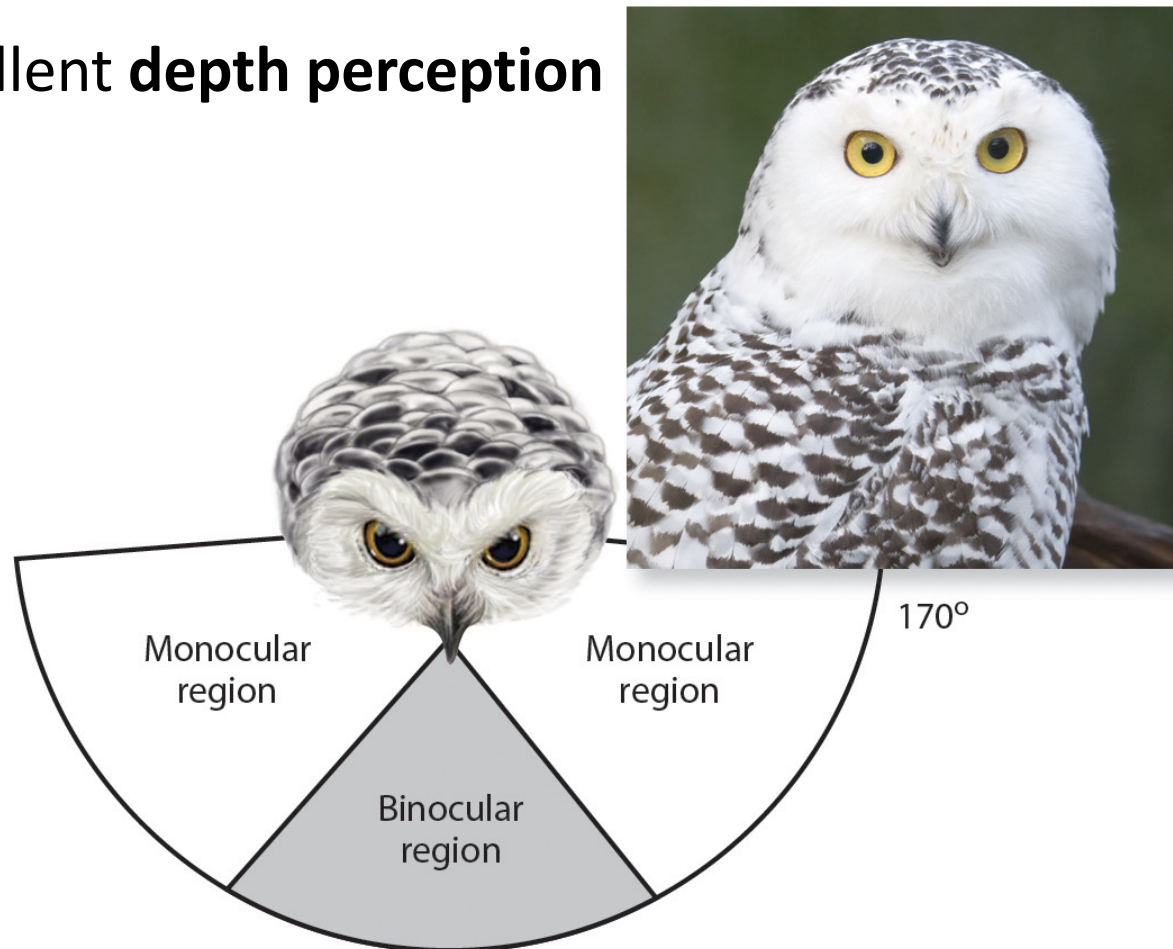
The retina refines the image

- Retina has **several layers of cells**
 - **Rods and cones** – deepest against sclera
 - **Pigmented epithelium** – absorbs light missed by photoreceptors
 - **Bipolar cells** – make synapses with photoreceptors and ganglion cells
 - **Ganglion cells** – send axons out of eye into optic nerve
- **Horizontal and amacrine cells** are interspersed across the retina
- **Horizontal cells** help define **boundaries** of an image
- **Amacrine cells** are important in **light adaptation** and sensitivity to movement
- Bipolar cells and ganglion cells are **transparent** so light can go through



Binocular vision

- Most vertebrates have **binocular vision**
 - Overlapping images are **elaborated in brain as one image**
 - Provides excellent **depth perception**

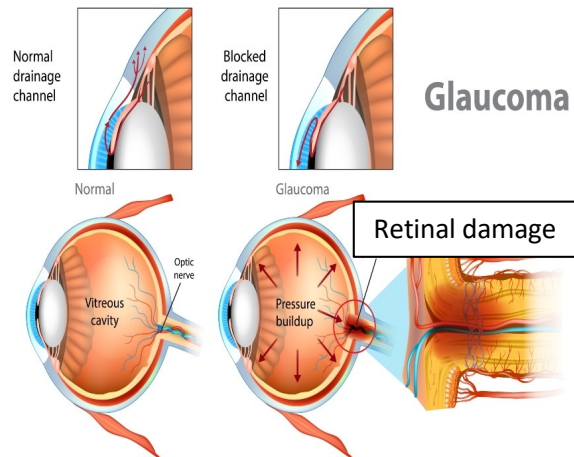


(a) Visual field of the snowy owl

Impact on Public Health: Visual disorders

—**Glaucoma** – drainage of **aqueous humor blocked** and **pressure inside eye rises**, causing general **damage of retina**

- **Less severe at the center** where high concentration photoreceptors
- 10% of all cases of blindness in US



—**Macular degeneration** – photoreceptor cells in macula **are lost**

- Causes loss of sharpness and color vision
- 25% of all cases of blindness in US

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a) How the world is seen by a person with glaucoma



b) How the world is seen by a person with macular degeneration

Impact on Public Health: **Visual disorders**

–**Cataracts** – accumulation of **protein clouds** the lens

- Lens can be removed and replaced
- 70% of people have cataracts by the age of 75

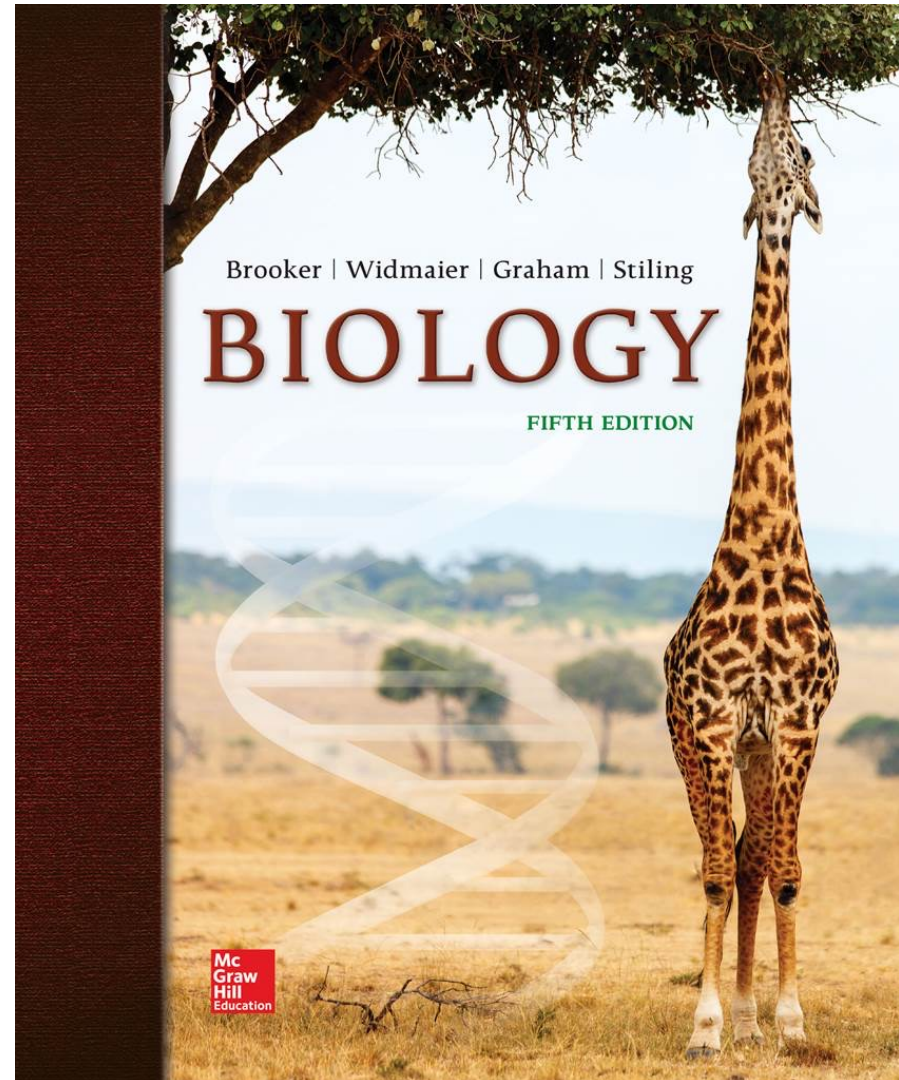


c) How the world is seen by a person with cataracts

Bibliography

Chapter 44 Lecture Outline

- See separate PowerPoint slides for all figures and tables pre-inserted into PowerPoint without notes.



Additional Bibliography

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by Fitzgerald Gruener Mtui

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ISBN-13: 978-0702037382

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Articles:

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